

Advantageous Selection in U.S. Transfer Programs*

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Abstract

Analyzing eight U.S. transfer programs, we show transfer recipients are highly negatively selected on consumption and lifetime income conditional on current annual income. Overall, transfers provide about 50 percent more to the consumption-poorest and lifetime-poorest than a negative income tax that is distributionally equivalent with respect to income. Using measures of simulated eligibility, we find selection primarily reflects take-up among the eligible rather than eligibility. We derive theoretical conditions under which these results imply welfare gains from transfers versus taxes. In our baseline calibration, approximately half of the welfare gains from tax-and-transfer reforms come from selection within income.

Keywords: transfer programs, take-up, eligibility, self-targeting, consumption inequality

JEL Codes: H23, H53, I38

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1 Introduction

In the United States, transfer programs that are neither universal entitlements nor fully taken up by those eligible for benefits play a major role in redistribution. In 2019, the bottom income quintile of households received transfers worth around 60 percent of their pre-tax income, according to estimates from the U.S. Congressional Budget Office. Yet the distinguishing features of transfers from income taxes—complex eligibility rules and voluntary take-up—suggest they may do something different from and complementary to tax-based redistribution. In particular, transfers may select households for assistance according to dimensions of need other than income.

The direction of selection into transfer receipt conditional on income, however, is unclear in practice. Economists have argued the purpose of both eligibility regulations and costly “ordeals” that depress take-up is to identify households whose needs are, by some standard, greater than other households (Nichols and Zeckhauser, 1982; Besley and Coate, 1992). Without such selection, the economic case for transfers relative to tax-based redistribution is perhaps questionable. Indeed, some scholars have concluded that transfers’ restrictive eligibility criteria and reliance on voluntary take-up risk perversely targeting benefits to less-needy households (Currie and Gahvari, 2008; Mullainathan and Shafir, 2013; Herd and Moynihan, 2019).¹ These perspectives disagree as to whether transfers actually succeed in selection and, more broadly, on the sign and magnitude of social welfare gains of targeting via eligibility rules and incomplete take-up.

In this paper, we first analyze selection into transfer receipt according to household consumption and lifetime income, long viewed as superior measures of living standards to annual income.² The eight programs we study constitute the vast majority of the U.S. safety net, costing about \$800 billion in total in 2019. To provide such an overview across programs, we use Panel Study of Income Dynamics (PSID) data on households’ incomes, consumption, and transfer receipt over time. We propose a selection test that governs our empirical analysis: Are transfer recipients needier than non-recipients with similar incomes? We then use a theoretical model of optimal redistribution to

¹For instance, Currie and Gahvari (2008) write that such burdens “often tend to exclude the neediest potential recipients because they are the ones who have the most difficulty complying with program rules.”

²See, e.g., Vickrey (1947). Auerbach et al. (forthcoming) and Fitzgerald and Moffitt (2022) are two recent examples.

interpret our findings. Both the empirical results and the model highlight the importance of what we call *within-income advantageous selection* to the social value of transfer programs: that the average transfer recipient has a lower living standard, and thus a higher marginal social welfare weight, than the average non-recipient with a similar income. Indeed, if transfers are sufficiently selective within income levels, then tax-and-transfer reforms may exist that simultaneously reduce consumption inequality and raise labor supply, even with an optimal nonlinear income tax.

We find within-income advantageous selection on both consumption and lifetime income for seven of the eight transfers we study, and this selection is usually economically significant in magnitude. Transfers therefore systematically select people whose living standards are lower than their incomes suggest. In the Supplemental Nutrition Assistance Program (SNAP), for instance, the average consumption rank of recipients is 18 percentiles lower than non-recipients with similar incomes, a difference of about \$5,100 per person per year or 35 percent of their average per-capita consumption. These gaps are slightly smaller for other transfers and for selection on lifetime income. Our interpretation is that transfers effectively distinguish temporarily low-income households, who can smooth consumption, from persistently low-income ones who cannot.

Why do transfers have these selection properties? One possible explanation is that eligibility rules may forbid receipt by low-income, high-consumption households. Another is that such households do not seek out transfers.³ We disentangle the contributions of eligibility and selective take-up among the eligible using newly-constructed measures of simulated eligibility for all eight transfers. We find that the primary force behind selection into transfers is take-up among the eligible, though households are typically less likely to be eligible when they have higher consumption, holding their incomes fixed. Among households that we classify as eligible for SNAP, for instance, the average consumption rank of recipients is 15 percentiles lower than that of non-recipients with similar incomes. Selection into SNAP among the eligible is thus only slightly weaker than selection among all households. Eligibility appears more important for some programs than others, but for

³There are many potential reasons for lower take-up among eligible high-consumption households, such as differences in welfare stigma, complementarity in consumption of in-kind transfers with inferior private goods, take-up costs that are amortized over long receipt spells, and differences in eligibility knowledge ([Anders and Rafkin, 2022](#)).

all transfers other than unemployment insurance, we can reject the null hypothesis of zero selection among the simulated-eligible into take-up on consumption or lifetime income given income. Given the well-known challenges of imputing transfer eligibility in survey data (e.g., [Duclos, 1995](#)), we carefully assess the sensitivity of our findings to the accuracy of these imputations.

Why should economists care about selection into transfers? We now turn to discussing its implications for real-world policy analysis and for social welfare. In terms of policy analysis, selection implies that transfers provide more resources to the consumption-poor and lifetime-poor than is suggested by their distributional incidence with respect to income. For households at the bottom of the distributions of consumption and lifetime income, we estimate that transfer programs provide about 50 percent more than a negative income tax that is distributionally equivalent in income terms. Another reference point for the incidence of transfers is that the selection effects we find entirely undo the effects of year-to-year income mobility and consumption-smoothing, which would otherwise make consumption and lifetime incidence less progressive than current incidence ([Poterba, 1989, 1991](#); [Fullerton and Lim Rogers, 1993](#)). The widespread use of income-based distributional analysis thus embeds a systematic understatement of transfer progressivity, reversing the typical critique of such analyses in the context of tax policy.

Our empirical evidence of conditional selection into receipt suggests that existing U.S. transfers may raise welfare as supplements to income taxation. To evaluate this claim, we develop a model of optimal redistribution with flexible tax and transfer schedules as functions of only current income. In our environment, transfers have incomplete take-up due to self-selection in response to application costs. The model extends the canonical [Nichols and Zeckhauser \(1982\)](#) environment to a more general setting suitable for welfare analysis. It yields formulas for the money-metric welfare effects of marginal budget-balanced reforms to taxes and transfers.

We use the model to study two potential “tax–transfer swaps” that incrementally shift the burden of redistribution between taxes and transfers, one of which leaves work incentives unchanged and another which is distribution-neutral with respect to income.⁴ The mechanism by which the

⁴The analysis presumes an optimal nonlinear income tax. This assumption is conservative, in that it isolates the comparison of taxes versus transfers as instruments from reforms that simply change overall redistribution.

government may raise welfare from a tax–transfer swap is that households with higher welfare weights may face lower take-up costs. If this correlation holds, then transfer take-up choices reveal welfare weights among sets of households that are indistinguishable to the income tax. In particular, if households with similar incomes but higher welfare weight also have lower take-up costs—that is, there is within-income advantageous selection—this force amplifies the [Nichols and Zeckhauser \(1982\)](#) motive to use transfers rather than taxes.

Using our empirical estimates to calibrate these welfare formulas, we draw two conclusions. First, the force of within-income advantageous selection is quantitatively important, providing about half of the welfare gains from transfers in our baseline calibration. This conclusion is, importantly, quite robust to assumptions about the social welfare function or other calibrated parameters. This robustness occurs because, for any given redistributive motive, consumption inequality within income is substantial, and transfers are well-targeted at this inequality. The second conclusion is that the strength of within-income selection is, in principle, large enough to produce overall welfare gains from increases in transfer benefit levels. For instance, an increase in the SNAP benefit level of one dollar for all households, financed by a lump-sum tax, produces nine cents of welfare gain per transfer dollar in our baseline calibration. 5.5 cents of the gain arise from within-income selection. On the other hand, some transfer expansions yield welfare losses. Sensitivity analyses find that the overall welfare effects of tax–transfer swaps are uncertain due to external parameters. The strength of within-income selection is often the deciding factor, as the other terms in the welfare formulas roughly cancel out.

Our analysis uses Panel Study of Income Dynamics (PSID) data from 1997 to 2019. The misreporting of transfer receipt and other key variables, namely income and consumption, poses in our view the most important challenge to our analysis ([Meyer et al., 2009, 2015](#)). We take these concerns seriously in several ways. First, on transfer misreporting, we adopt the multivariate corrections of a recent literature that estimates how misreporting probabilities vary with household characteristics by linking survey and administrative data ([Davern et al., 2019](#); [Mittag, 2019](#); [Meyer et al., 2020](#)). These corrections actually strengthen our results. Second, on consumption

misreporting, our results hold for consumption subcategories thought to be well-measured, as well as for most measures of durable goods ownership, following [Meyer and Sullivan \(2023\)](#). Third, on lifetime income, we extend methods from [Haider and Solon \(2006\)](#) to address potential biases from incomplete income histories, and the comparison to independent measures of consumption offers a form of replication ([Caspersen and Metcalf, 1994](#)). Fourth, on transfer eligibility, our results are robust to reclassifying simulated-ineligible recipients as eligible and to comparisons of observably-similar households, which are likelier to have the same true eligibility status ([Duclos, 1995](#)). While our analysis ultimately relies on survey data, our battery of sensitivity analyses gives us some confidence that our main results are not artifacts of mismeasurement.

Our paper advances several literatures in public finance. Our primary contribution is to the literature on the take-up of transfers. Following [Nichols and Zeckhauser \(1982\)](#) and [Besley and Coate \(1992\)](#), many theoretical analyses have considered the importance of “self-targeting” to the efficiency of in-kind transfers (as reviewed in [Currie and Gahvari, 2008](#)). As an empirical matter, the existence and sign of these selection forces has been an active area of recent research (e.g., [Mullainathan and Shafir, 2013](#); [Deshpande and Li, 2019](#); [Herd and Moynihan, 2019](#); [Finkelstein and Notowidigdo, 2019](#); [Lieber and Lockwood, 2019](#); [Homonoff and Somerville, 2021](#)). We propose and implement a selection test that is simple, portable across programs, and maps naturally into a welfarist framework.⁵ Our test most closely resembles an analysis in [Alatas et al. \(2016\)](#) showing that hassle costs in an Indonesian transfer program deter applications from people with unobservably high consumption, controlling for an observable proxy used in eligibility determinations.

We also contribute to a descriptive literature on distributional incidence analysis. Several recent papers examine the distributional incidence of taxes and transfers in other countries from a lifetime perspective ([Blundell et al., 2015](#); [Bengtsson et al., 2016](#); [Roantree and Shaw, 2018](#); [Brewer et al., 2020](#); [Levell et al., 2021](#)). Our analysis focuses on the U.S. and examines incidence with respect

⁵Previous papers on the welfare effects of ordeals examine the causal effect of a specific ordeal and thus identify their selection properties on the margin. In contrast, we quantify the average level of selection into transfer receipt, which aggregates across all forces that induce selection, including ordeals, stigma, eligibility rules, and other forms of self-selection. Moreover, even if specific barriers to take-up affect between-income selection into transfers, research has seldom examined transfers’ within-income selection properties.

to both consumption and lifetime income.⁶ The most similar paper in this regard is [Auerbach et al. \(forthcoming\)](#), which studies U.S. fiscal progressivity from a lifetime perspective. Our analysis builds on theirs by identifying the central role of selective take-up in shaping the consumption and lifetime incidence of transfers. A recent paper on student loan forgiveness, [Catherine and Yannelis \(2023\)](#), also finds striking differences between incidence concepts, which like ours occur because of selection into receipt on lifetime income conditional on current income.

Finally, we contribute to a large theoretical literature on transfers. First, we provide a novel formalization and generalization of the [Nichols and Zeckhauser \(1982\)](#) problem, along with empirically implementable measures of welfare effects of marginal reforms to transfers. Along the way, we provide a natural sufficiency condition under which distributions of take-up costs can yield efficiency gains from tax–transfer swaps. To prove these results, we employ a version of the [Jacquet and Lehmann \(2014\)](#) perturbation approach. Our distribution-neutral tax offsets echo the welfare concepts in [Diamond and Sheshinski \(1995\)](#), [Kaplow \(2010\)](#), [Hendren \(2020\)](#), and others.

2 Data and Measurement

Our main source of data is the Panel Study of Income Dynamics (PSID) in its eleven biennial survey waves from 1997 to 2019. In each PSID wave, we observe heads of household and spouses ages 18 to 65. We first review key aspects of the data, leaving further details to Appendix B. We then explain three imputation procedures that augment the PSID data: for cash-equivalent values of in-kind transfers, transfer eligibility, and lifetime income.

The PSID data have positive and negative aspects. Our analysis would be mostly infeasible in U.S. data other than the PSID, which uniquely has a long panel dimension as well as measures of income, consumption, and receipt of many transfers. Nevertheless, the data come with significant

⁶[Fullerton and Metcalf \(2002\)](#) reviews the literature on the lifetime incidence of taxation. The few related U.S. studies of lifetime transfer incidence often assume away the take-up margin of transfers, treating them as equivalent to taxes ([Davies et al., 1984](#); [Fullerton and Lim Rogers, 1993](#); [Guner et al., 2021](#)). Of course, other strands of research imply that selective take-up could be important for distributional analysis of transfers: Scholars have found heterogeneous income dynamics among low-income households ([Bane and Ellwood, 1986](#)) and selective take-up among the eligible ([Blank and Ruggles, 1996](#); [Currie, 2006](#); [Ganong and Liebman, 2018](#); [Finkelstein and Notowidigdo, 2019](#)).

reporting issues that we discuss in depth in Section 3. We conclude there that our results appear mostly robust to known issues with misreporting in survey data.

2.1 Income, Consumption, and Transfer Receipt

Current Income. We define household income as total annual income of the head and spouse before taxes and transfers, excluding other household members due to data constraints. Income includes labor, business, and capital income. Following the National Academy of Sciences (Citro and Michael, 1995), we adjust for household size using the equivalence scale $e_h = (N_{h,\text{adult}} + 0.7N_{h,\text{child}})^{-0.7}$, where $N_{h,\text{adult}}$ and $N_{h,\text{child}}$ respectively denote the numbers of adults and of children in household h .⁷

Current Consumption. The PSID has extensive coverage of consumption expenditures since 1999. Expenditure categories covered include food, housing, health care, transportation, education, child care, and several smaller topics. We make two adjustments to the data to better measure consumption rather than expenditure, following Meyer and Sullivan (2023). First, for homeowners, we replace mortgage and property tax payments with equivalent rents based on reported home values. Second, for vehicle owners, we replace loan payments with estimates of lease-cost equivalents. Household consumption is then equivalized by the same scale as income.

Transfer Receipt. We observe self-reported household receipt for nine transfer programs. These are the Supplemental Assistance Nutrition Program (SNAP); Medicaid; Section 8; public housing; Temporary Assistance for Needy Families (TANF); Supplemental Security Income (SSI); unemployment insurance (UI); Women, Infants, and Children (WIC); and the Low Income Home Energy Assistance Program (LIHEAP). Due to extensive overlap in the modern operations of U.S. housing programs, we combine public housing and Section 8 into a single program to which we refer as “housing assistance,” and results distinguishing between them are in Appendix A.

⁷Appendix A includes results not adjusted for household size and composition. These are typically quite similar to those with equivalized households. By implication, other equivalence scales are also likely to yield similar results.

2.2 Imputation of Other Variables

Valuing Transfers. We measure the dollar value of transfers by combining information from the PSID and the Supplemental Poverty Measure module of the U.S. Current Population Survey (CPS). For SNAP, TANF, SSI, UI, and LIHEAP, the PSID records the nominal value of transfers over various time periods, which we rebase as the per-capita annualized amount in 2020 constant dollars. The PSID does not include cash-equivalent values for in-kind transfers, namely Medicaid, Section 8, public housing, and WIC. We impute these amounts with the average values by household size and year reported in the CPS.

Lifetime Income. We construct a lifetime concept of household income from incomplete income histories. To begin, we estimate a Poisson regression model with individual fixed effects, interacted with age-specific coefficients as recommended by [Haider and Solon \(2006\)](#). Letting i index individuals, t index calendar years, and a index age in years, the model takes the following form:

$$E[y_{it} | X_{it}] = \exp(\alpha_i \lambda_a + X'_{it} \beta_a), \quad (1)$$

where α_i is an individual fixed effect, α_t is a year fixed effect, X_{it} is a matrix of time-varying demographic characteristics, and λ_a and β_a are vectors of age-specific coefficients. The outcome y_{it} is individual income.

We then perform four adjustments before using the regression results to estimate lifetime income. First, we apply empirical Bayes methods from [Morris \(1983\)](#) to shrink the estimates $\hat{\alpha}_i$ of the individual fixed effects towards a conditional expectation fit from time-invariant individual characteristics. Second, in an outer loop, we project estimates $\hat{\alpha}_i$ on income y_{it} , using Equation 1, to obtain age-specific coefficients $\hat{\lambda}_a$, iteratively re-estimating $(\hat{\alpha}_i, \hat{\lambda}_a)$ until convergence.⁸ After convergence, we balance the panel, imputing demographic characteristics as from their nearest observed values, and we then calculate lifetime average income from ages 18 to 65. Fourth, we account for spousal income in a way that permits changes in household composition over time.

⁸We initialize $\lambda_a = 1$ and thus estimate the canonical fixed-effect model of lifecycle income in the first step.

Let $j(i, t)$ indicate i 's spouse in year t . Our concept of lifetime household income follows each individual through the sequence of households they experience as adults. That is, the lifetime household income of individual i is

$$\bar{y}_i^h = \sum_t e(\hat{y}_{it}^h) = \sum_t e(\hat{y}_{it} + \hat{y}_{j(i,t),t})$$

where t is again summed over the years in which i is between ages 18 and 65, and $e(\cdot)$ is the equivalence-scale function. If we restrict our sample to stable households (as in, e.g., [Fullerton and Lim Rogers, 1993](#)), our definition of lifetime income would coincide with the standard concept.

Simulated Eligibility. We simulate transfer eligibility in the PSID by compiling information on program rules, mainly from primary-source documents and research databases of such rules, similar to the Urban Institute's TRIM program ([Zedlewski and Giannarelli, 2015](#)). See Appendix B for details on these eligibility simulations.

Simulated-eligibility measures cannot perfectly measure true eligibility, as there are differences between information used in actual eligibility determinations versus what is recorded in surveys. In validation checks in Appendix B, we confirm that our simulated eligibility measure is highly predictive of benefit receipt, though misclassification is apparent. Significant fractions of recipients are simulated to be ineligible, and take-up rates are counterfactually low among those simulated to be eligible. Both are routine issues in eligibility simulations ([Duclos, 1995](#)). The main threat from mismeasured eligibility in our analysis is that we may understate the importance of eligibility rules in determining the distributional incidence of transfers. In sensitivity analyses, we show how our results change when we reclassify simulated-ineligible recipients as eligible, which mechanically raises the influence of eligibility rules relative to take-up.

3 Understanding Selection into Transfer Receipt

This section presents our main empirical results. We quantify the magnitude of “within-income advantageous selection”: Holding incomes fixed, we find that transfer recipients have lower con-

Table 1: Average Annual Per-Capita Total Transfer Benefits
by Quintile of Current Income, Lifetime Income, and Consumption

		Current Income Quintile				
		1	2	3	4	5
Consumption Quintile	1	3,516	1,567	868	773	138
	2	2,346	1,262	556	343	239
	3	1,924	974	587	390	212
	4	1,396	952	624	417	159
	5	967	864	666	368	154
		1	2	3	4	5
Lifetime Income Quintile	1	3,323	1,521	820	657	311
	2	2,399	1,275	746	457	174
	3	2,013	1,178	607	410	212
	4	1,728	1,080	636	394	184
	5	1,807	1,049	581	358	141

Notes: This table reports the average annual per-capita total value of transfer benefits, cash and in-kind, by quintiles of equivalized household current income, lifetime income, and consumption. Values are in constant 2020 dollars.

sumption and lifetime income than non-recipients. Within-income advantageous selection into receipt is mostly explained by take-up among the eligible and not eligibility. Finally, we assess the sensitivity of our results to measurement issues.

3.1 Transfer Tabulations

Table 1 reports the average annual per-capita value of transfer benefits for households at a given combination of income and consumption quintiles. Moving up the income distribution, households unsurprisingly receive less in transfers, consistent with advantageous between-income selection. Yet, comparing households in the same income quintile but which have different levels of consumption, lower-consumption households receive substantially more in transfers than higher-consumption households—advantageous within-income selection. For instance, among households in the bottom income quintile, those also in the bottom consumption quintile receive more than three times as much in transfers as those also in the top consumption quintile. Similar results appear for lifetime income.

Is this pattern likely to be explained by the eligibility rules of transfer programs, by selective

Table 2: SNAP Receipt, Eligibility, and Take-Up Rates by Income and Consumption Quintile

Panel A: Receipt Rate

		Income Quintile				
		1	2	3	4	5
Consumption Quintile	1	53.0	23.1	7.8	4.1	3.3
	2	31.1	15.2	4.0	1.8	0.5
	3	20.1	7.9	3.3	0.7	0.4
	4	9.8	7.0	2.5	0.5	0.1
	5	7.1	4.0	1.5	0.5	0.1

Panel B: Simulated Eligibility Rate

		Income Quintile				
		1	2	3	4	5
Consumption Quintile	1	82.8	28.5	0.4	0.2	0.0
	2	76.8	22.8	0.4	0.1	0.0
	3	71.1	20.0	0.4	0.2	0.1
	4	68.1	20.1	0.6	0.3	0.1
	5	62.1	25.0	1.0	0.5	0.1

Panel C: Take-Up Rate Among Simulated Eligibles

		Income Quintile				
		1	2	3	4	5
Consumption Quintile	1	54.5	36.1	.	.	.
	2	33.5	30.2	.	.	.
	3	24.3	17.6	.	.	.
	4	12.5	15.3	.	.	.
	5	9.9	9.1	.	.	.

Notes: This table reports the shares of households that receive SNAP (Panel A), are simulated to be eligible for SNAP (Panel B), and take up SNAP conditional on being simulated to eligible (Panel C). Households are split by quintile in the distributions of equivalized household consumption and income. Due to low rates of simulated eligibility, we do not report take-up rates for the top three income quintiles. See Appendix A for a tabulation by income and lifetime income.

take-up among the eligible, or by some combination of these two forces? Table 2 sheds some initial light on this question, taking SNAP as an example. Panel A shows that, consistent with Table 1, receipt rates of SNAP are falling in both income rank and consumption rank given income rank. Panel B shows rates of simulated eligibility for SNAP by income and consumption quintile. Unsurprisingly, SNAP eligibility falls quickly in income, and there are few households above the second income quintile that are SNAP-eligible. The rate of simulated SNAP eligibility is also falling in consumption given income but less markedly than does receipt. What explains the difference

between these panels is the take-up rate among simulated eligibles, which is shown in Panel C. Take-up rates fall sharply in consumption given income. Among eligible households in the bottom income quintile, the SNAP take-up rate among those also in the bottom consumption around 55 percent, as compared to approximately 10 percent among top-consumption-quintile households.⁹ Overall, Table 2 suggests that selection into SNAP on consumption given income is explained primarily by the take-up behavior among the eligible and secondarily by eligibility.

From the perspective of public-policy analysis, a key consequence of these selection patterns is that transfers provide more resources to the consumption-poor and lifetime-poor than their incidence with respect to current income would suggest. Income-based distributional analyses are pervasive in policy settings but are typically thought to overstate the progressivity or regressivity of policies, as year-to-year household income fluctuations make consumption and lifetime incidence more uniform than current incidence (Poterba, 1989, 1991). In the context of transfers, however, selection into receipt would mitigate the mobility effect.

To show the implications of such selection for distributional analysis, we estimate receipt rates and average dollar values of benefits per person per year as locally-linear functions of households' percentile rank in a given distribution. We rank households on the basis of their equivalized current income, current consumption, and lifetime income.¹⁰ Panel A of Figure 1 plots the average total annual per-capita value of transfer benefits as functions of these ranks. In the left sub-panel, we combine transfers across the eight transfer programs we study. The right sub-panel excludes in-kind programs, for which we must impute cash-equivalent values of benefits. Panel B plots estimates of receipt rates as a function of ranks for the eight transfer programs we study.

Both panels also show that transfer incidence with respect to consumption, lifetime income, and current income are remarkably similar, overall and for most individual transfer programs.

⁹While the *levels* of transfer take-up rates are sensitive to the general expansiveness or conservativeness of any simulation of eligibility, measurement issues seem less likely to explain the vast *differences* in take-up rates by consumption given income. Intuitively, one requires dramatically higher underreporting rates of receipt by low-income, high-consumption households or extensive misclassification of such households as transfer-eligible—a theme to which we return in Section 3.3.

¹⁰Appendix A contains supplementary figures that probe the sensitivity of our results to definitional choices and provide detail on distributional incidence.

That is, average annual per-capita benefits are about as high at the bottom of the distributions of consumption and lifetime income as they are at the bottom of the current-income distribution. Such results are perhaps surprising because of the aforementioned compressive effect of income mobility, which mechanically reduces the lifetime or consumption incidence of transfers. This effect is fully undone by selection into receipt. We present this point explicitly with the dashed lines, which apply the empirical transition matrix from income ranks to consumption ranks to the current incidence of transfers.¹¹ Furthermore, the similarity of consumption and lifetime incidence suggests transfer programs identify households with low consumption as a result of persistently low income, rather than a lesser ability to smooth consumption relative to income over time.¹²

3.2 Regression Analysis of Selection

Using rank–rank regressions, we now estimate whether transfer-recipient households have lower consumption or lifetime-income ranks than households with similar incomes who do not receive transfers. This approach isolates selection and summarizes it over the entire joint distribution of consumption and income. In particular, our regression specification is

$$\bar{R}_{it} = \alpha_{ct} + \beta D_{it} + f(R_{it}) + \Omega_{it} + u_{it}, \quad (2)$$

where $\bar{R}_{it}$ is the consumption rank or lifetime-income rank for person i in year t , R_{it} is i 's current-income rank, $f(R_{it})$ is a flexible function of this rank, D_{it} indicates i 's receipt status for a given transfer program, and α_{ct} is a cohort–year fixed effect. In some specifications, we restrict the sample to people who are simulated-eligible and include demographic controls Ω_{it} . The coefficient β summarizes the extent of within-income selection into a transfer.

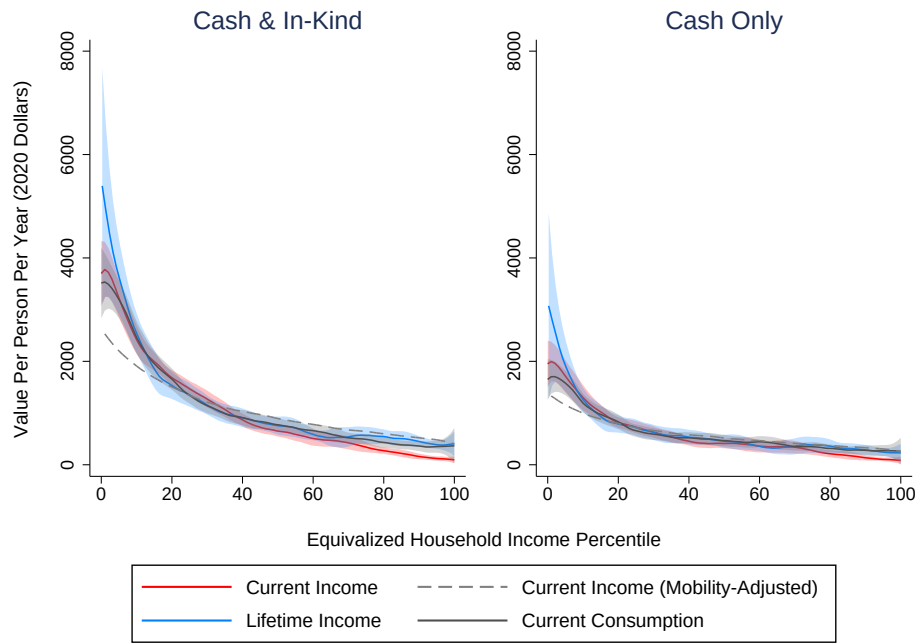
Figure 2 shows that transfer receipt is associated with a lower consumption rank and lifetime-income rank, conditional on current-income rank (blue circles). Across programs, transfer recipi-

¹¹As an example, for the receipt rate $r(y)$ at current-income rank y , this mobility counterfactual is $r(c) = \int r(y) dF(y|c)$, where $F(y|c)$ is the conditional distribution of income rank at consumption rank c .

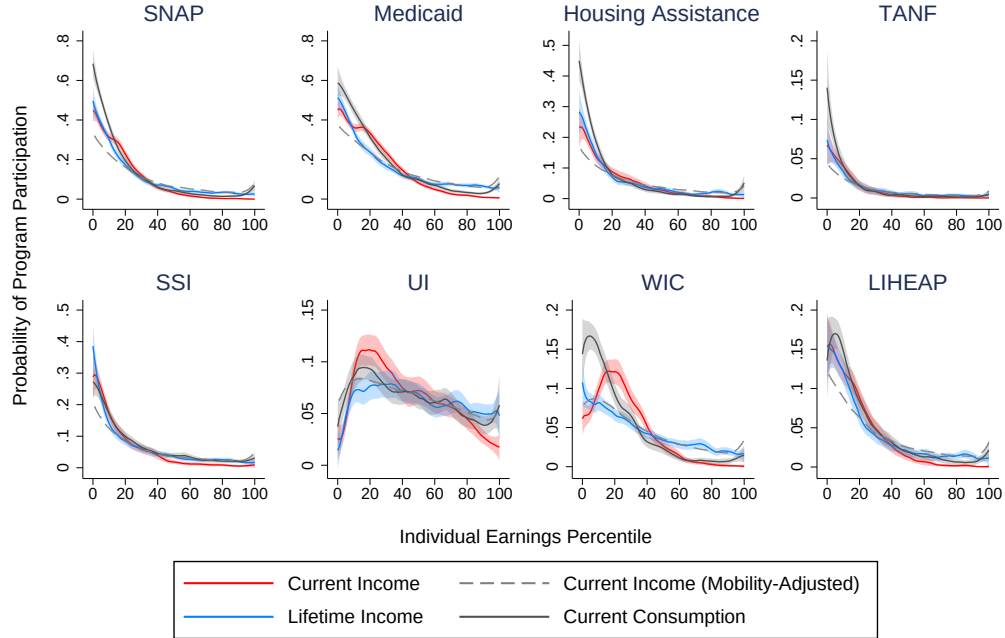
¹²Caspersen and Metcalf (1994), one of the few other studies to analyze both consumption incidence and lifetime incidence, also find that they are quite similar to each other.

Figure 1: Receipt and Value of Transfer Benefits as a Function of Equivalized Household Rank

Panel A: Average Total Annual Per-Capita Benefits



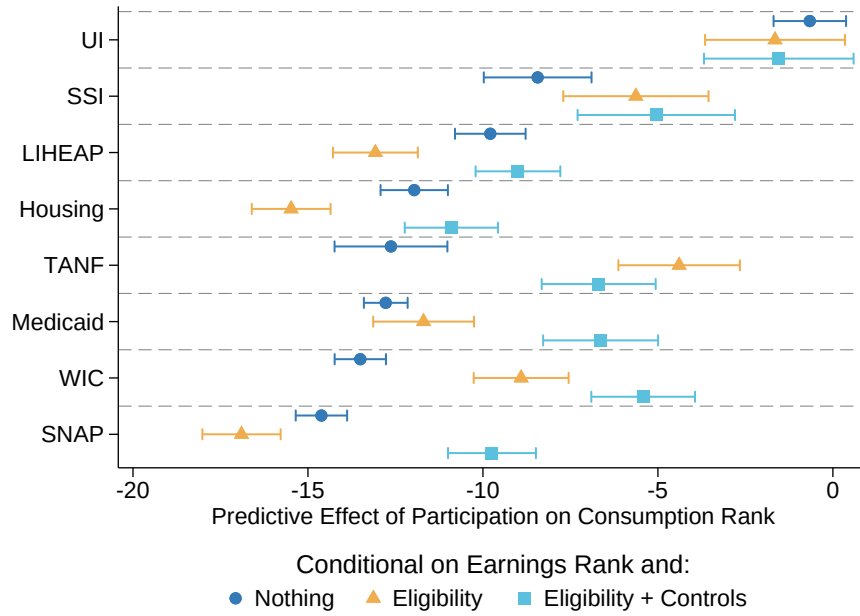
Panel B: Receipt Rate by Program



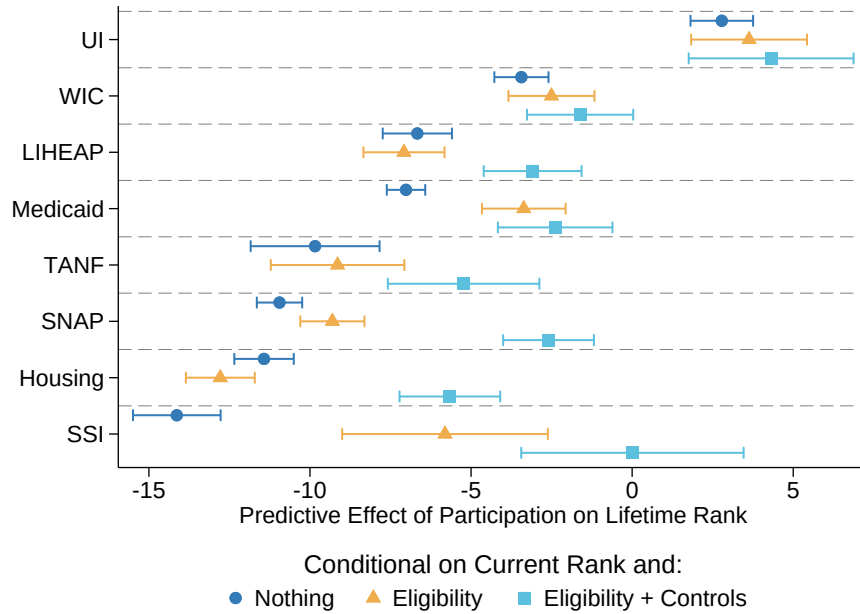
Notes: This figure displays average total annual per-capita values of benefits and receipt rates by program as functions of household ranks in the distributions of equivalized current income, lifetime income, and current consumption. The functions are estimated by local linear regressions with bandwidths of three percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in [Chernozhukov et al. \(2013\)](#), with clustering by individual.

Figure 2: What Explains Selection into Transfer Receipt?

Panel A: Selection on Consumption



Panel B: Selection on Lifetime Income



Notes: This figure displays estimates of the predictive effect of transfer receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient γ from Equation 2). For estimates represented by blue circles, we do not add additional control variables to the specification, whereas for the yellow diamonds, we estimate the regression only on people whom we simulate to be eligible, and for the teal squares, we subset to the simulated-eligible population and control for a rich set of demographic variables. The confidence intervals are for the 95-percent level and reflect clustered standard errors by individual.

ents rank on average 10 to 15 percentiles lower in the consumption distribution than non-recipients of that program with similar current incomes.

The extent of selection into transfers on consumption and lifetime income varies considerably across transfer programs. For instance, SNAP and Medicaid are highly informative about consumption and lifetime income, whereas UI is essentially uninformative about consumption given income. The difference of UI from other transfers underscores the practical difference between contributory and non-contributory social insurance. Some “non-differences” are also interesting: for example, it is not the case that cash-like transfers (e.g., SNAP) are systematically more selective than the transfers most unlike cash (e.g., housing or Medicaid). In a few cases, there are marked differences between selection on consumption and selection on lifetime income. Most notably, SSI recipients are highly negatively selected on lifetime income but are much less selected on consumption.

Limiting the sample to the simulated-eligible does little, on average, to reduce the average gaps in consumption and lifetime income between recipients and non-recipients (yellow triangles). Most of the selection into transfer receipt on consumption given income therefore appears to reflect selective take-up among the eligible rather than eligibility rules. Mismeasurement of eligibility mostly appears unlikely to explain this conclusion. When we reclassify any simulated-ineligible recipients of a given program to be eligible for that program, we do not see a clearer role for eligibility in concentrating incidence among the consumption-poor (Appendix Figure A3). The transfer program where our results are potentially most sensitive to mismeasurement of eligibility is SSI. Even with this adjustment to simulated eligibility, take-up remains the primary force in shaping the consumption and lifetime incidence of transfers.

3.3 Sensitivity to Mismeasurement

Survey data are imperfect. Here we consider whether known issues in reported transfer receipt, income, and consumption could explain our results. We also address potential concerns with our measures of lifetime income and simulated transfer eligibility.

Transfer Receipt. Using linked survey and administrative data, [Mittag \(2019\)](#) and [Davern et al.](#)

(2019) estimate statistical models of household survey reporting behavior for SNAP and Medicaid respectively. Their models, intended for use as misreporting corrections, predict the probability of true transfer receipt given survey-reported receipt and demographic characteristics. Intuitively, these models allow researchers to replace assumptions of constant misreporting rates with misreporting probabilities that are functions of demographic observables.

Their corrections consistently increase our estimates of selection (Appendix Table A1). There are two reasons why. First, under constant misreporting rates, our estimates are attenuated. Consider misreporting probabilities $p_0 = \Pr(\tilde{D}_i = 0 | D_i = 1)$ and $p_1 = \Pr(\tilde{D}_i = 1 | D_i = 0)$, where D_i indicates true receipt and $\tilde{D}_i$ indicates reported receipt. Comparing the feasible regression of $y_i = \tilde{\beta}\tilde{D}_i + u_i$ to the infeasible regression $y_i = \beta D_i + u_i$, we have that $\beta = \tilde{\beta}/(1 - p_0 - p_1)$.¹³ Second, the parameter estimates in Mittag (2019) and Davern et al. (2019) both imply that underreporting of transfer receipt is somewhat more common among households with low consumption and lifetime income, holding income constant. Thus, their adjustments tend to amplify the increase in the selection that we would find under constant misreporting rates.

As a distinct exercise, we also calculate the rates of underreporting at the top of the consumption distribution that would be necessary to yield zero selection among the eligible (Appendix B). Overturning our conclusions requires a degree of underreporting that we view as implausible, such as “false-negative” rates of 50 percent in the top quarter of the consumption distribution. Though misreporting of transfer receipt is an important phenomenon, it is unlikely to explain the selection patterns in this paper.

Transfer Eligibility. We have already reclassified simulated-ineligible recipients as eligible to probe the sensitivity of our results to mismeasurement of eligibility. Another check in this spirit is to estimate the extent of selection into transfers, as in Equation 2, with controls so that we compare recipients and non-recipients who are observably similar and thus are likelier to have the same true eligibility status.¹⁴ Adding these controls eliminates some but not all of the remaining selection

¹³Meyer et al. (2009) finds rates of under-reporting rates in the PSID in the range of 7 to 27 percent across programs we study. This suggests a presumption that our main estimates in Figure 1 are understated, even for transfer programs where heterogeneous-misreporting corrections have not yet been estimated.

¹⁴We condition on covariates that are used in eligibility rules for at least one transfer program. These are the

into receipt for most programs (Appendix Figure ??). An improved measure of eligibility is thus unlikely to overturn a substantial role for take-up in selection into receipt.

Income and Consumption. Income is poorly measured at the bottom of the distribution, and while there is also mismeasurement in consumption, it appears less severe than for income (Meyer and Sullivan, 2003; Brewer et al., 2017). How is mismeasurement likely to affect our results?

One story of intentional misreporting likely pushes in the opposite direction of our results. That is, transfer recipients have incentives to underreport income to maintain eligibility. Such incentives apply less strongly to consumption, and to nonrecipients. All else equal, transfer recipients would thus appear *positively* selected on consumption given income. This is opposite to what we find.

To address concerns about measurement error in consumption, we show in Appendix Table A2 that similar selection patterns appear when only looking at categories of consumption deemed “well-measured” in Meyer and Sullivan (2023). For instance, Medicaid recipients consume 36 percent less in housing, 25 percent less in vehicles, and 24 percent less in food at home than similar-income people who are not on Medicaid. Similar patterns also manifest in PSID data on durable-goods ownership, such as whether the household owns a home, car, computer, and the number of rooms or presence of air conditioning in the home (Meyer and Sullivan, 2012), as we show in Appendix Table A3. SNAP recipients are 16 percentage points less likely to own a home, 13 percentage points less likely to own a car, and 12 percentage points less likely to own a computer than similar-income people not on SNAP. The consistency of selection across measures, some of which seem truly unlikely to suffer from meaningful mismeasurement, bolsters our findings.

Lifetime Income. Inferring lifetime income from “snapshots” is challenging (Haider and Solon, 2006). If one thought that mismeasurement of lifetime income for people with fewer years-in-sample imparts a systematic bias in our results, then the estimated extent of selection into transfers should “drift” up or down as one examines selection for households with more or fewer years-in-sample. We re-estimate the predictive effects of transfer receipt on lifetime rank as in Equation

household’s state of residence, household size and composition, family income and earnings, ages of household members, disability status, unemployment duration and reason, and basic measures of wealth (value of any automobiles and liquid assets).

2, splitting individuals by years-in-sample. We show in Appendix Figure A4 that selection is remarkably stable once we observe someone for at least two survey waves.

3.4 Extensions

Decomposition. Appendix A presents methods and results for an accounting decomposition of the roles of income mobility, eligibility, and take-up in shaping the consumption incidence and lifetime incidence of transfers relative to their current incidence. The comparison of dashed lines in Figure 1 to actual consumption incidence is, in fact, the mobility effect in Brewer et al. (2020). We extend their decomposition by distinguishing between the contributions of eligibility and take-up. Consistent with Figure 2, our decomposition results suggest a dominant role for selective take-up in concentrating incidence at the bottom of the distributions of consumption and lifetime income.

Inverse Analysis. Inverting the analysis of Figure 1, we estimate the income-tax reforms necessary to replicate the consumption and lifetime incidence of transfers. We show in Appendix Table A5 that the required reforms are far more progressive in income than the incidence of transfers with respect to income, due to the selection properties of transfers. To replace all transfer programs, the average marginal tax rate in the bottom tenth of the income distribution would need to increase by about 25 percentage points. That increase is roughly twice the “tax” rate implied by the slope of receipt in income apparent in Figure 1. The exercise is static and does not account for behavioral responses to policy changes.

Selection Over Time. Our data span 1997 to 2019, allowing us to address how the U.S. safety net has evolved over time. We estimate a version of Equation 2 that allows for year-specific coefficients on transfer receipt. To allow us to describe broad trends, we “stack” the data over programs and include program-specific controls. Across our eight transfer programs, we see little change in selection on consumption over time, but a considerable intensification of selection on lifetime income (Appendix Figure A6).

4 Welfare Analysis of Transfers

What do our results imply about optimal transfer policy? This section considers an optimal-redistribution problem in which the government can implement both nonlinear tax and transfer schedules. The aim of the theory is to derive conditions under which a marginal increase in the transfer benefit level, financed by an income-tax increase, raises social welfare. We then calibrate the theoretical propositions using PSID data to provide rough estimates of the welfare effects.

4.1 Model Setup

Household's Problem. Each household has a multidimensional type $\theta = (\delta, \kappa)$ distributed according to $\mu(\theta)$. The parameter δ is the household's productivity, and κ is their cost of applying to the transfer. There are no income effects on labor supply, so each productivity level maps onto a unique wage w . Households supply labor to generate pre-tax income z , which is then taxed according to the non-linear schedule $T(z)$, and the remainder is consumed.

There is also a transfer with nonlinear schedule $S(z)$, to which households apply and receive if their benefit exceeds their take-up cost. However, to ensure no bunching and well-defined elasticities, we assume households choose their labor supply before they see their realization of κ .¹⁵ As shorthand, we use $\mathbb{1}_S = S(z) \geq \kappa$ as an indicator for whether, ex post, the individual applies for and receives the transfer. We think of households as choosing their pre-tax income z directly. To rule out income effects, we assume households have quasi-linear utility in cash and cash-equivalent transfer dollars. Specifically, each household solves the program:

$$V(\theta) = \max_z \left\{ z - T(z) - v(z/w) + \int_0^{S(z)} (S(z) - \kappa) \mu(w, \kappa) d\kappa \right\}. \quad (3)$$

The household's optimal choice z^* implies their ex-post consumption is $c = z - T(z) + \mathbb{1}_S S(z)$.

Government's Problem. The government chooses tax and transfer schedules $T(\cdot)$ and $S(\cdot)$ to

¹⁵Here we abstract away from eligibility rules, which can be viewed as affecting take-up costs.

maximize utility summed across individuals according to type-specific Pareto weights ($\alpha(\theta)$):

$$\max_{T,S} \int_{\Theta} \alpha(\theta) V(\theta) d\mu(\theta),$$

subject to a balanced-budget constraint:

$$\int_{\Theta} [T(z(\theta)) - \mathbb{1}_S \times S(z(\theta))] = 0$$

and to consumer optimization. In what follows, we use the social marginal welfare weights $g(\theta) = \alpha(\theta) V'_c / \lambda$, where λ is the marginal value of public funds, to concisely define optimal taxes and transfers. As is usual in this setting, we assume that social marginal welfare weights decrease with income, all else equal: $\frac{\partial}{\partial z} g(\theta) < 0$.

Our results are delineated by two distinct relationships between social marginal welfare weights and underlying type parameters.

Definition 1. *We say that take-up is advantageously selected within income when $\frac{\partial}{\partial \kappa} g(\theta) < 0$.*

Within-income advantageous selection means that, all else equal, households with low application costs to the transfer program have higher marginal welfare weights.¹⁶ That is, the households that are first to apply for the transfer at any income level have high marginal welfare weights. If recipients are negatively selected on consumption given income (as in Section 3), advantageous selection holds, assuming social welfare weights that decrease in consumption. On the other hand, if ordeals perversely screen out needier people (e.g., [Herd and Moynihan, 2019](#)), within-income advantageous selection does not hold.

We define $M(z)$ and $m(z)$ as respectively the cumulative distribution and density functions of transfer application costs for households at income z : $M(z) = \Pr(S(z) \geq \kappa)$ for all those earning income z , and similarly $m(z) = \Pr(S(z) = \kappa)$. We assume the distributions are continuous on $\kappa \geq 0$, that is, there are no mass points. These distributions are endogenous to tax and transfer

¹⁶Definition 1 follows [Saez and Stantcheva \(2016\)](#) in abstracting away from any specific rationale.

schedules, and for notational brevity, we suppress the dependence on $S(z)$. Of particular interest is the case when the mass of inframarginals to the program is monotonic in income.

Definition 2. *We say households are advantageously selected between income when the distribution of costs κ conditional on income z increases in z in the sense of first-order stochastic dominance.*

Between-income advantageous selection evokes [Nichols and Zeckhauser's \(1982\)](#) targeting argument for transfers. If take-up costs increase in z , then take-up rates are declining in income, and hence transfers will have positive redistributive effects between incomes.

4.2 Policy Reforms

We evaluate the welfare effects of transfers under this setup by contemplating two policy reforms. In both, we offer all households a flat increase of ds in the benefit level $S(z)$ at all incomes z . The reforms are budget-neutral: the government pays for benefit increases with differing tax changes. Thus, we sometimes call these reforms “tax–transfer swaps,” because they raise transfers and taxes simultaneously while holding the total budgetary impact fixed.

Reform 1. To offset the transfer increase of ds , the planner imposes a lump-sum income-tax increase of $\tau = \int_z [dsM(z) + S(z)m(z)] h(z)dz$ at all incomes z . The increase offsets the static cost of the increased benefit to the inframarginals and fiscal cost from marginal recipients. Because this reform imposes lump-sum changes to both the transfer and income tax, it has no efficiency effects via labor supply.

Reform 2. To offset the transfer increase of ds , the planner increases income taxes by $\tau(z) = M(z)ds$ at each income z , plus a flat tax of $E_z [(S(z) + ds)m(z)]$ to cover the fiscal cost from marginal recipients. Marginal rates thus change by $\tau'(z) = \frac{d}{dz}M(z)ds$ at z . The purpose of this reform is to be budget-neutral within income z , and thus non-redistributive between incomes. The reform generally induces a labor-supply response. The reform then redistributes any increase in revenue due to the labor-supply response via lump-sum transfers.

In the next proposition, we decompose the welfare effects of each reform:

Proposition 1. *The welfare effects of Reforms 1 and 2 are respectively*

$$\begin{aligned}
\Delta W_1 = & \underbrace{ds \int_z M(z) (E_{\kappa \leq S(z)} [g(z, \kappa)] - E_{\kappa} [g(z, \kappa)]) h(z) dz}_{\text{redistribution within } z} \\
& + \underbrace{ds \int_z E_{\kappa} [g(z, \kappa)] (M(z) - E_{z'} [M(z')]) h(z) dz}_{\text{redistribution between } z} \\
& - \underbrace{E_z [(S(z) + ds)m(z)] E_{z, \kappa} [g(z, \kappa)]}_{\text{fiscal cost of marginals}}.
\end{aligned} \tag{4}$$

and

$$\begin{aligned}
\Delta W_2 = & \underbrace{ds \int_z M(z) (E_{\kappa \leq S(z)} [g(z, \kappa)] - E_{\kappa} [g(z, \kappa)]) h(z) dz}_{\text{redistribution within } z} \\
& + \underbrace{ds \int_z M'(z) z \frac{T'(z) - \frac{d}{dz}(S(z)M(z))}{1 - T'(z)} E_{\kappa} [\epsilon^z] E_{z', \kappa} [g(z', \kappa)] h(z) dz}_{\text{labor supply effect}} \\
& - \underbrace{E_z [(S(z) + ds)m(z)] E_{z, \kappa} [g(z, \kappa)]}_{\text{fiscal cost of marginals}},
\end{aligned} \tag{5}$$

where $\epsilon^z = \frac{\partial z}{\partial \tau'} \frac{1 - T'(z)}{z}$ is the elasticity of income with respect to a small change τ' in the marginal tax rate of those with initial income z .¹⁷

We now discuss the implications of these welfare formulas. Under what conditions do these tax–transfer swaps raise welfare? How do the welfare effects change as the initial benefit level rises? Is it ever possible to reduce inequality and raise labor supply via a tax–transfer swap? Proofs of all propositions are in Appendix C.

In our setting, unlike in Nichols and Zeckhauser (1982), the first transfer dollar cannot raise efficiency. This is because the benefits and costs of the first dollar are second-order for all recipients, an implication of the envelope theorem. Welfare effects of the first dollar arise entirely from the

¹⁷In Appendix C, we give conditions under which the elasticity of income is properly defined.

labor-supply response and fiscal cost of the higher tax schedule. But starting at an optimal tax schedule, these effects—which run solely through tax changes—must net to exactly zero or else the tax system cannot have been initially optimal. A welfare effect of the first dollar can be recovered if some households face zero or negative take-up costs or do not initially optimize. Alternatively, if the planner faces constraints on imposing the optimal income tax, then the first dollar of transfers may raise welfare relative to the constrained optimum.

Proposition 2. *After the first transfer dollar, the welfare effect of within-income redistribution is positive if there exists within-income advantageous selection. The welfare effect of between-income redistribution is positive if there exists both within- and between-income advantageous selection.*

Once there are inframarginals $M(z)$, then advantageous selection is sufficient for the transfer to have positive redistributive effects. However, the signs of ΔW_1 and ΔW_2 depend also upon the fiscal cost of marginal recipients. The relative importance of within- and between-income redistribution in Reform 1, or within-income redistribution and the labor-supply effect in Reform 2, depend upon the nature of selection into receipt and the shape of $g(\theta)$.

The next proposition addresses how the welfare gains from redistribution change as the transfer expands further.

Proposition 3. *Given advantageous selection, the welfare effects from within- and between-income redistribution both have an inverted U shape: They are initially small, have a maximum at some finite benefit level, and then diminish.*

The proposition establishes that there is a limit to potential welfare gains from transfers versus taxes. Marginal welfare effects are initially increasing because early transfer dollars mostly offset take-up costs, whereas later dollars hit a rising share of inframarginals. Eventually, the strength of advantageous selection must deteriorate, as take-up by households with low productivity δ or low take-up cost κ becomes exhausted.

Our final proposition addresses the potential for efficiency gains, not only redistributive gains, from tax–transfer swaps.

Proposition 4. *Suppose the tax system is optimal and we have between-advantageous selection. Then the labor supply effect in Equation (5) is positive.*

Proposition 4 connects our framework to [Nichols and Zeckhauser \(1982\)](#). If higher-income households have lower take-up rates, then uniform increases in the benefit levels are progressive. The tax change that recoups the average transfer at each income level z is thus regressive and raises more revenue at low incomes. Consequently, marginal tax rates fall everywhere, possibly allowing for an efficiency gain. In particular, assuming advantageous selection both within and between incomes, a tax–transfer swap exists with positive redistributive and labor-supply effects under an optimal income tax. However, signing the ultimate welfare effect also requires estimating the fiscal effect of marginal recipients.¹⁸

4.3 Quantifying the Welfare Effects of Transfers

The above formulas suggest the potential for welfare gains from transfers. It remains an empirical question whether welfare gains emerge at plausible parameter values, how economically large they are, and the relative importance of between- versus within-income advantageous selection in these welfare effects. Tying together our theoretical and empirical analyses, we now calibrate the formulas using our PSID estimates and several external inputs.

We provide these estimates with caution. We calibrate the nature of the social welfare function, the take-up and labor-supply elasticities, and the slope of the tax/transfer schedules, all of which are subject to considerable uncertainty. The sign of the effects of the policy reforms depends on these calibrations, so economists can reasonably disagree. However, the importance of within-income redistribution as a share of the total redistributive effect is robust across calibrations.

Calibration. Our PSID data yield estimates of receipt rates $M(z)$, average benefits $S(z)$, and the income distribution $h(z)$. We calibrate the elasticity of taxable income ϵ^z at 0.3, the midpoint of the range of estimates in [Saez et al. \(2012\)](#). We use a twice-differentiable approximation to the

¹⁸Our condition echoes [Diamond and Sheshinski \(1995\)](#): “Having [the receipt rate] $p(\theta)$ increasing in [disability] θ is sufficient to make a disability program desirable, provided the marginal utility of consumption of non-workers exceeds that of workers at the optimum without a disability program.”

tax schedule from [Heathcote et al. \(2017\)](#), $T(z) = y - \alpha y^{1-\beta}$, and their estimate for β of 0.151, a parameter that controls the progressivity of taxes.¹⁹ The parameter α is chosen to match [Heathcote et al.’s \(2017\)](#) average marginal tax rate of 33 percent. These steps yield $T(z)$ and its derivatives. The remaining two parameters, the welfare weights $g(\theta)$ and the density of application costs $m(z)$, we construct by combining PSID and external inputs as follows.

In our primary analysis, we assume a utilitarian social planner and households who have CES utility over consumption, characterized by risk aversion parameter γ . We calibrate $\gamma = 3$ from [Chetty and Finkelstein \(2013\)](#) and show robustness to this choice. In a robustness check, we use inverse-optimum welfare weights ([Hendren, 2020](#)) that rationalize the existing tax schedule. Welfare weights in hand, we use empirical transfer receipt rates by income and consumption in the PSID to construct the empirical analog of $E_{\kappa \leq S(z)} [g(z, \kappa)]$, which is the average welfare weight among transfer recipients at income level z . Finally, we estimate the density of application costs $m(z)$ so as to match the estimated take-up elasticity with respect to the benefit level of 0.5 in [Krueger and Meyer \(2002\)](#). In particular, for an elasticity η of take-up with respect to the benefit level, $m(z) = \eta M(z)/S(z)$.

Results. Table 3 reports our primary estimates of the welfare effects of the tax–transfer swaps outlined above. Columns 1–4 apply to Reform 1, reflecting the three terms in Equation 4, and Columns 5–8 apply to Reform 2 and therefore reflect the terms in Equation 5. We report results for each transfer and for a dollar-weighted average of the transfers.

We step through the interpretation of Table 3, taking SNAP as an example. Reform 1 produces welfare gains from redistribution of about 14 cents per dollar increase in the transfer. Of these gains, 40 percent (5.5 cents out of 13.9) come from redistribution among households that have similar current incomes but different consumption levels. These welfare gains reflect the strong within-income advantageous selection of SNAP that we document in Section 3.

¹⁹Notice that this specification implies that a ten-percent increase in pre-tax income yields an increase in post-tax income of 8.49 percent ($1 - 0.151 = 0.849$). Under a constant marginal tax rate, $\beta = 1$.

Table 3: Welfare Effects of Tax–Transfer Swaps (Cents per Transfer Dollar)

	Reform 1: Lump-Sum Tax Increase				Reform 2: No Redistribution Between Incomes			
	(1) Redistribution Within	(2) Redistribution Between	(3) Fiscal Effect of Marginals	(4) <i>Total</i>	(5) Redistribution Within	(6) Labor Supply	(7) Fiscal Effect of Marginals	(8) <i>Total</i>
Dollar-weighted average	1.6	4.2	-4.0	1.9	1.6	2.3	-4.0	-0.1
SNAP	5.5	8.4	-4.9	9.0	5.5	2.5	-4.9	3.1
Medicaid	2.3	8.8	-7.2	3.9	2.3	3.9	-7.2	-1.0
Housing Assistance	5.5	4.3	-2.6	7.2	5.5	1.2	-2.6	4.2
TANF	0.9	1.0	-0.5	1.4	0.9	0.2	-0.5	0.5
SSI	2.2	5.0	-2.9	4.4	2.2	1.3	-2.9	0.7
UI	-0.7	0.8	-3.1	-3.0	-0.7	2.1	-3.1	-1.8
WIC	-0.8	1.7	-1.9	-1.0	-0.8	0.7	-1.9	-2.0
LIHEAP	1.2	3.2	-1.9	2.5	1.2	0.9	-1.9	0.2

Notes: This table reports estimates of the welfare effects of two types of budget-neutral reforms that jointly adjust tax and transfer schedules. Both reforms make all households eligible for an additional transfer dollar. Reform 1 pays for this transfer expansion via a lump-sum tax on all households, whereas Reform 2 changes income tax rates so as to exactly net out any redistribution between incomes. All columns report the money-metric welfare gains in cents per transfer dollar. Columns correspond to the terms of Equations 4 and 5, where we divide each term by the average welfare weight to yield a money-metric interpretation.

Another 8.3 cents of the welfare gains come from downward redistribution between income levels. These gains emerge because the SNAP take-up rate is decreasing with respect to income, so a SNAP benefit increase financed by a lump-sum tax is progressive with respect to income. Three cents of welfare gains per transfer dollar are eroded by fiscal costs of marginal recipients. Altogether, the net effect is a welfare gain of nine cents per transfer dollar. Recall that these tax-transfer swaps are budget-neutral, so the nine-cent increase comes simply from manipulating tax and transfer schedules.

There are two important conclusions from Reform 1. First, the welfare effects of within-income redistribution are typically large as a share of total redistribution (Column 1 divided by the total of Columns 1 and 2). Said differently, within-income advantageous selection plays a substantial quantitative role in the overall redistributive motive for transfers. Second, the welfare effects of benefit increases differ considerably by program. Expanding SNAP with Reform 1 yields large welfare gains, while expanding UI and WIC with Reform 1 would reduce welfare. Differences across transfers in the degree of within-income selection correspond tightly to differences in overall welfare effects of Reform 1.

Next, we examine Reform 2, which eliminates the role of redistribution between incomes. Across programs, Reform 2 always generates a positive labor supply effect. This effect is an immediate consequence of the fact that we find positive between-income redistribution (Column 2) and the result in Proposition 4, which says that between-income redistribution and the appropriate reform can engineer efficiency gains. The magnitude of the labor-supply effect is similar or smaller than the fiscal effect of marginals. As a result, the size and sign of the welfare effect of Reform 2 typically depends on the size of within-income redistribution. For several large programs like SNAP and Housing Assistance, Reform 2 induces positive welfare effects. But the dollar-weighted average impact of the reform is economically small and negative, owing to the large negative effects of Medicaid, UI and WIC. Altogether, welfare-increasing tax-transfer swaps intended to induce positive labor supply effects are difficult to produce across programs, but such reforms could exist in several programs.

Sensitivity Analysis. We make many stark assumptions in the above analysis. In Table 4, we show that the importance of within-income redistribution is remarkably robust to different assumptions.

We first vary the risk-aversion parameter γ of the social welfare function (Rows 2–5). When we change risk aversion to 1 or 8, the welfare gains from redistribution within incomes remain 33–46 percent as large as the gains from between-income redistribution. By contrast, welfare gains from between-income redistribution are not robust to the strength of risk aversion (i.e., how egalitarian is the social welfare function). We also find a large role for within-income redistribution if we posit utility over lifetime income rather than consumption. In Row 5, we use inverse-optimum social welfare weights from Hendren (2020).²⁰ This is the specification in which we find within-income redistribution is least important relative to between-income redistribution. This occurs because, as Hendren (2020) notes, the inverse-optimum welfare weights suggest the U.S. does not meaningfully prioritize the poor versus the middle class on the margin.

Table 4 also suggests that, without more confidence in the external inputs, it is hard to say whether there are welfare gains overall from tax–transfer swaps. If the social planner places less value on redistribution, then the redistributive effects do not outweigh the fiscal cost. On the other hand, with more value on redistribution, the dollar-weighted average welfare effect of Reform 2 can be positive in some specifications (Column 8). Moreover, varying the calibrated take-up elasticity and elasticity of taxable income has sizable effects (Rows 6–7) which depend linearly on the respective elasticity.

In summary, the new channel we identify in this paper—within-income advantageous selection—appears to play an important quantitative role in the welfare analysis of transfers. However, the sign of the overall welfare effect of transfer expansions is quite sensitive to considerations beyond this paper.

²⁰The weights are given over current income, so we then solve for the welfare weight schedule over consumption that matches these inverse-optimum weights using a Tikhonov regularization procedure (Appendix B).

Table 4: Welfare Effects of Tax–Transfer Swaps, Robustness

	Reform 1: Lump-Sum Tax Increase				Reform 2: No Redistribution Between Incomes			
	(1) Redistribution Within	(2) Redistribution Between	(3) Fiscal Effect of Marginals	(4) <i>Total</i>	(5) Redistribution Within	(6) Labor Supply	(7) Fiscal Effect of Marginals	(8) <i>Total</i>
1. Primary dollar-weighted estimate	1.6	4.2	-4.0	1.9	1.6	2.2	-4.0	-0.2
<i>Varying the Social Welfare Function</i>								
2. SWF curvature $\gamma = 8$	2.5	5.5	-4.0	4.0	2.5	2.2	-4.0	0.7
3. SWF curvature $\gamma = 1$	0.6	1.8	-4.0	-1.6	0.6	2.2	-4.0	-1.3
4. SWF over lifetime income	2.7	8.4	-4.0	7.1	2.7	2.2	-4.0	0.9
5. Inverse-optimum SWF (Hendren, 2020)	0.3	1.5	-4.0	-2.2	0.3	2.2	-4.0	-1.5
<i>Varying Calibrated Elasticities</i>								
6. Take-up elasticity $\eta = 0.3$	1.6	4.2	-2.4	3.5	1.6	2.2	-2.4	1.4
7. ETI $\varepsilon^z = 0.1$	1.6	4.2	-4.0	1.9	1.6	0.7	-4.0	-1.6

Notes: This table reports estimates of the welfare effects of two types of budget-neutral reforms that jointly adjust tax and transfer schedules. Both reforms make all households eligible for an additional transfer dollar. Reform 1 pays for this transfer expansion via a lump-sum tax on all households, whereas Reform 2 changes income tax rates so as to exactly net out any redistribution between incomes. All columns report the money-metric cents of welfare gains per transfer dollar. Columns correspond to the terms of Equations 4 and 5, where we divide each term by the average welfare weight to imply a money-metric interpretation. See Appendix B for details about computing Row 5.

Caveats. One way in which these calculations stop short of a complete cost–benefit analysis of transfer programs is that they do not account for changes in the government’s administrative costs. It seems likely that transfers are administratively more costly than tax-based redistribution. [Isaacs \(2008\)](#), for instance, estimates that Medicaid spends about 5 cents per benefit dollar, SNAP about 16 cents per benefit dollar, and WIC at least 27 cents per benefit dollar. By comparison, the EITC’s administrative cost is estimated at 1.5 cents per dollar. Applying these cost estimates in our setting, however, seems unsound. Little is known about the extent to which administrative costs of transfers are fixed versus variable in their number of recipients or total transfer dollars. While the estimated welfare gains from transfers seem likely to be offset in part by administrative-cost differentials, the existing evidence is not strong enough to pin down the relative magnitudes.

This analysis has several other limitations. First, we ignore the insurance value of fiscal progressivity ([Stepner, 2019](#)). Second, we ignore behavioral responses to transfers beyond labor supply. Two examples of such responses are fiscal impacts of cross-program enrollment spillovers and dynamic incentives for human-capital investment. Third, we assume homogeneous labor supply and take-up elasticities. Heterogeneity in elasticities could shift the results in either direction. For instance, non-recipients may have lower take-up elasticities than recipients, which would reduce the fiscal cost from marginal recipients. Third, we make the strong assumption that households optimize, despite a large and growing literature documenting departures from classical behavior in benefit take-up ([Bhargava and Manoli, 2015](#); [Finkelstein and Notowidigdo, 2019](#); [Anders and Rafkin, 2022](#)). If households do not initially optimize, that can generate an additional positive welfare effect from benefit take-up of marginal enrollees, since they would no longer be subject to an envelope condition.

5 Conclusion

It has long been understood that taxes on annual personal income can struggle to address disparities in lifetime resources and thus in living standards ([Vickrey, 1947](#)). This paper argues that, in the U.S., transfers systematically correct for this limitation of income taxes. We document this pattern

empirically, and we interpret it theoretically in a model of optimal redistribution.

Our key empirical contributions are to show that transfer receipt is highly negatively selected on consumption and lifetime income given income and that this selection primarily reflects take-up among the eligible rather than eligibility rules. Our main theoretical contribution is to provide formulas that quantify the welfare gains from transfers, in particular by imagining two “tax–transfer swaps” that incrementally shift the burden of redistribution from taxes to transfers. Calibrating these formulas using our empirical results, we find within-income redistribution is of central importance to the welfare analysis of transfers, contributing about half of the welfare gains from budget-balanced reforms that increase benefit levels.

The core message of this paper is that advantageous selection into transfers means that incomplete take-up is, from a welfarist perspective, more of a blessing than a curse. Eliminating such selection also eliminates the social value of transfers when the government levies an optimal income tax. In particular, the power of transfers to detect needier households within income levels represents a novel, quite general, and quantitatively significant source of their welfare gains.

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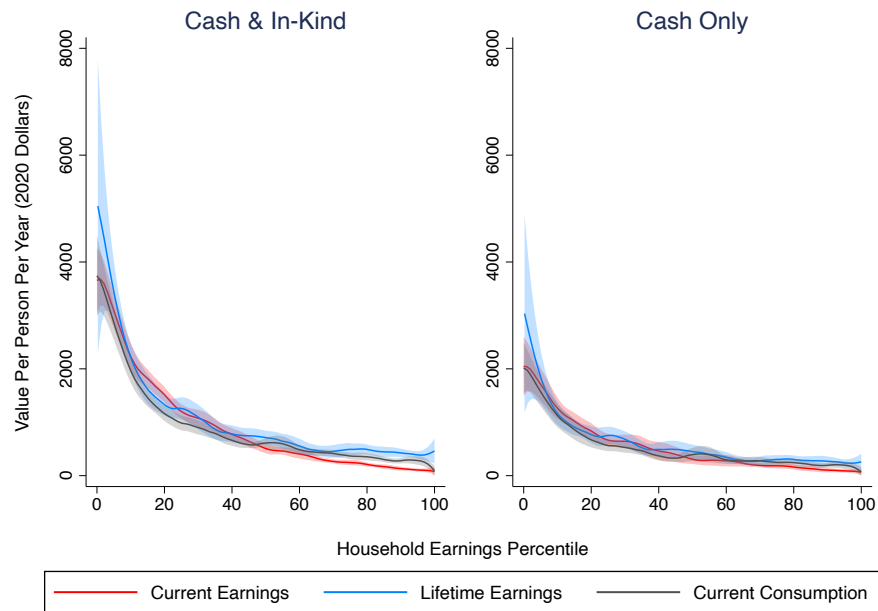
Appendices for Online Publication

A	Additional Tables and Figures	39
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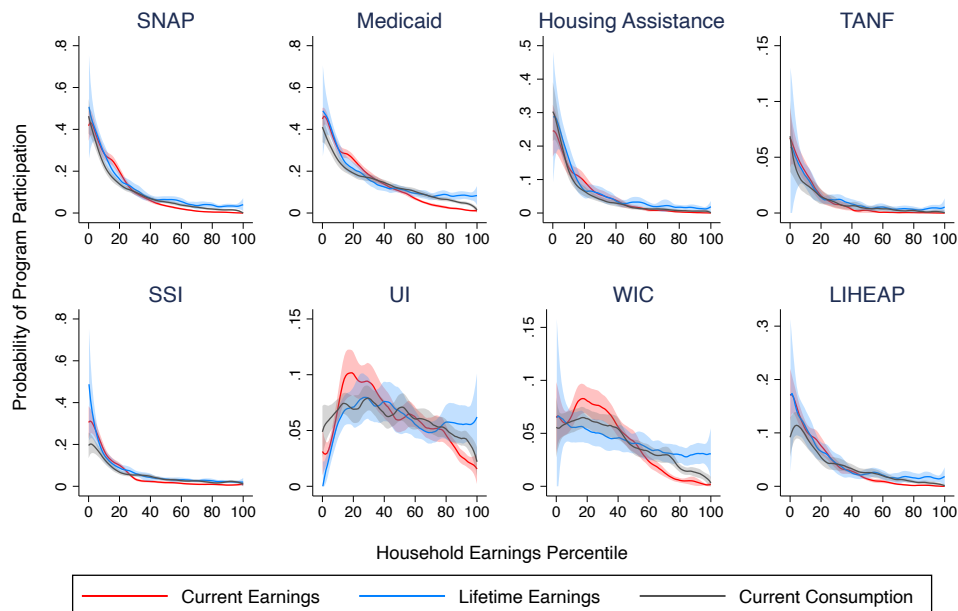
A Additional Tables and Figures

Figure A1: Receipt and Value of Transfer Benefits as a Function of Household Rank

Panel A: Average Total Annual Per-Capita Transfer



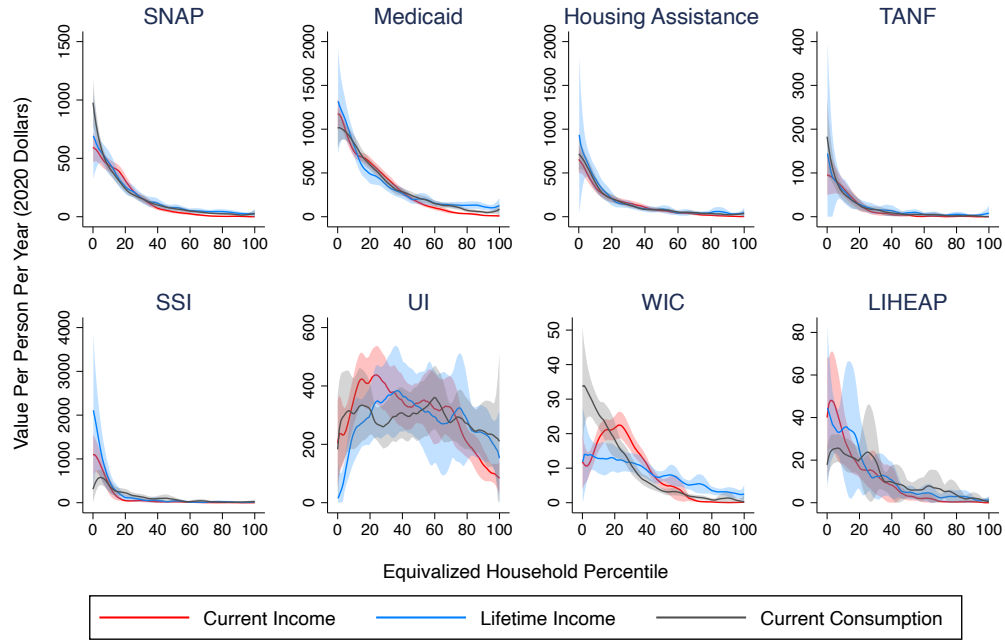
Panel B: Receipt Rate by Program



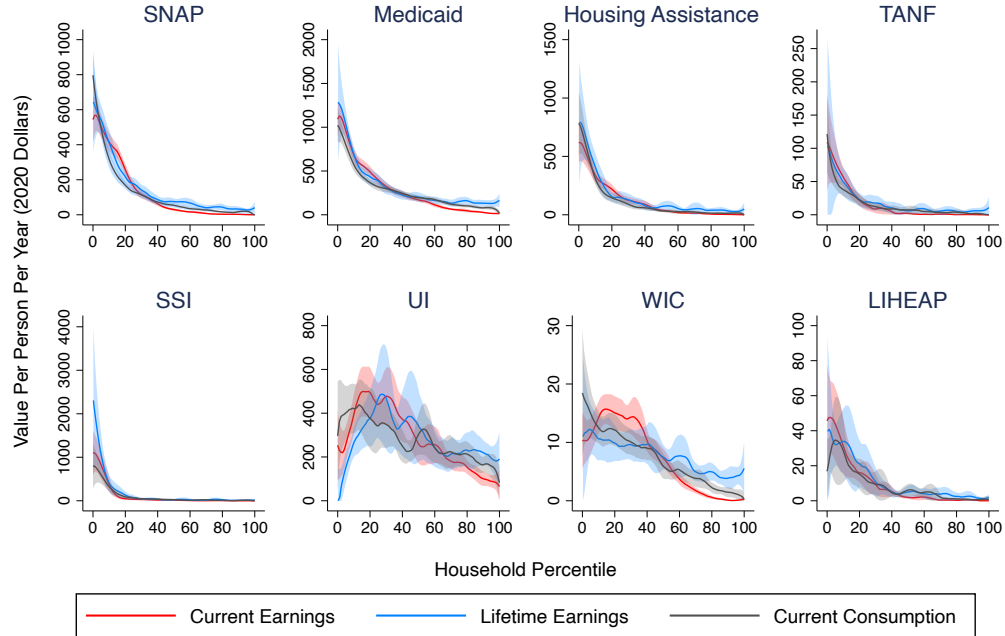
Notes: This figure displays average total annual per-capita values of benefits and receipt rates by program as functions of household ranks in the distributions of household current income, lifetime income, and current consumption. The functions are estimated by local linear regressions with bandwidths of three percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in [Chernozhukov et al. \(2013\)](#), with clustering by individual.

Figure A2: Value of Transfer Benefits, by Program, as a Function of Rank

Panel A: Equivalized Household Rank

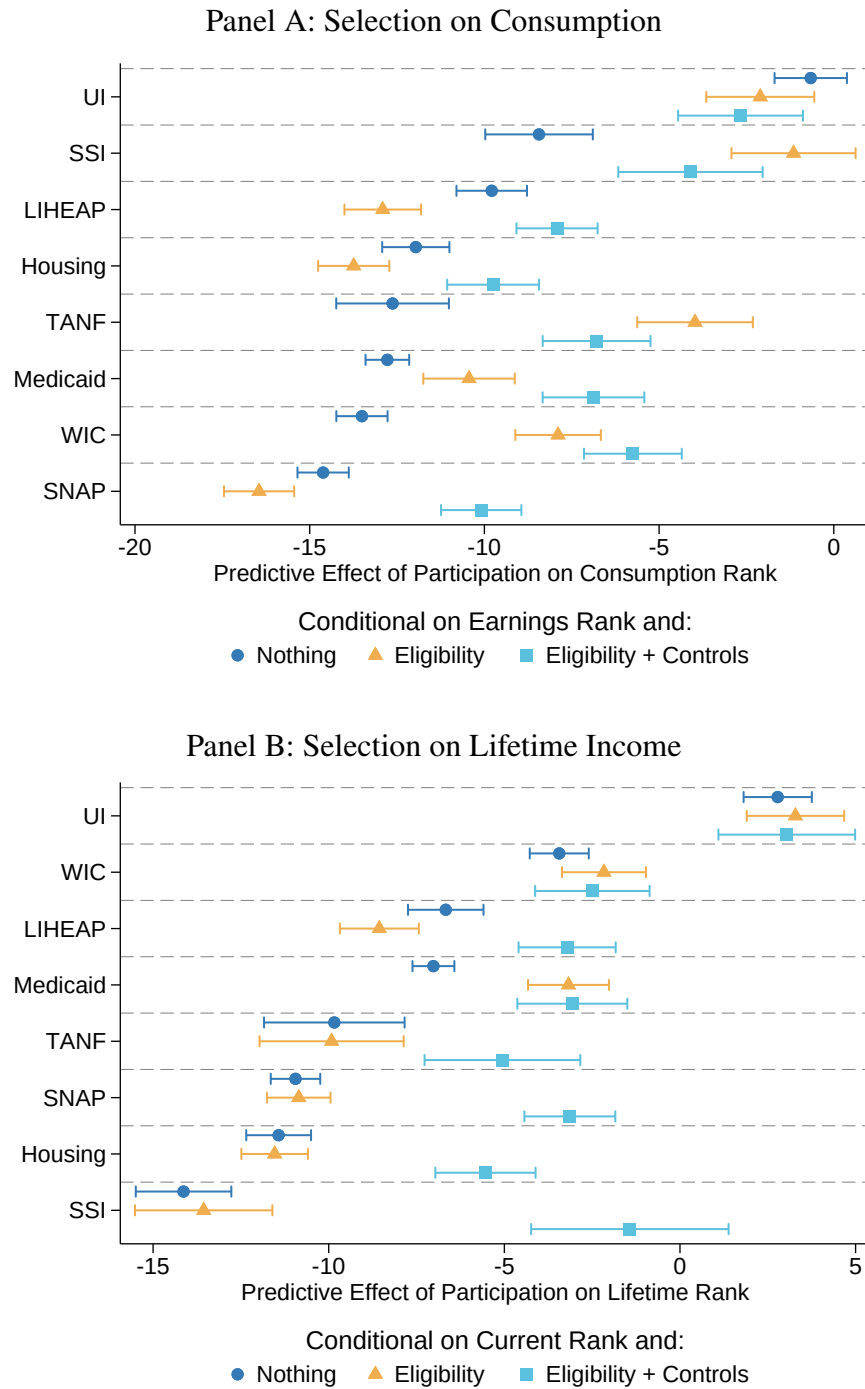


Panel B: Household Rank



Notes: This figure displays the average total annual per-capita value of transfer benefits by program as functions of ranks in the distributions of current income, lifetime income, and current consumption. Panel A reports results for equivalized income or consumption, whereas Panel B does not adjust for differences in household size and composition. The functions are estimated by local linear regressions with bandwidths of three percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in [Chernozhukov et al. \(2013\)](#), with clustering by individual.

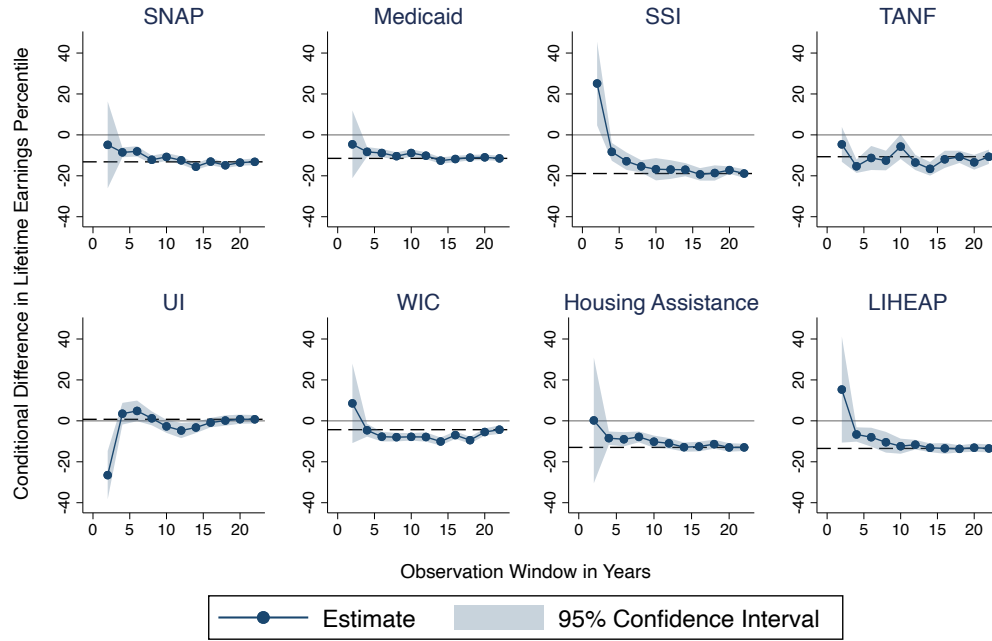
Figure A3: What Explains Selection into Transfer Receipt?: Reclassifying Simulated Eligibles



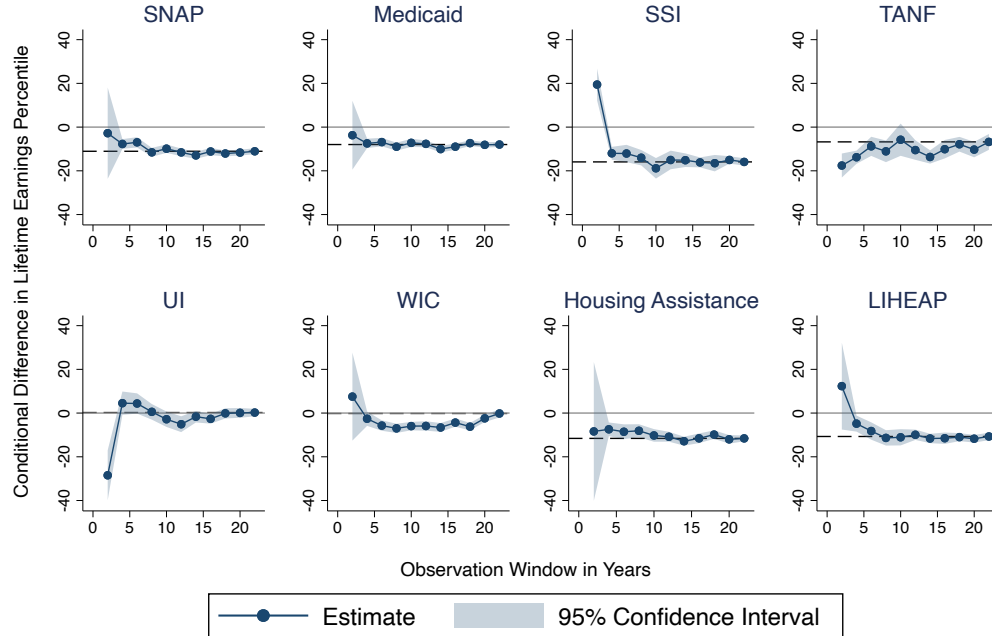
Notes: This figure displays estimates of the predictive effect of transfer benefit receipt on consumption rank or lifetime-income rank, conditional on current-income rank (coefficient γ from Equation 2). For estimates represented by blue circles, we add no additional control variables to the specification, whereas for the yellow diamonds, we add program eligibility, and for the teal squares, we add eligibility and a rich set of demographic controls. The confidence intervals are for the 95-percent level and clustering by indi.

Figure A4: Transfer Receipt as a Function of Rank

Panel A: Equivalized Household Rank

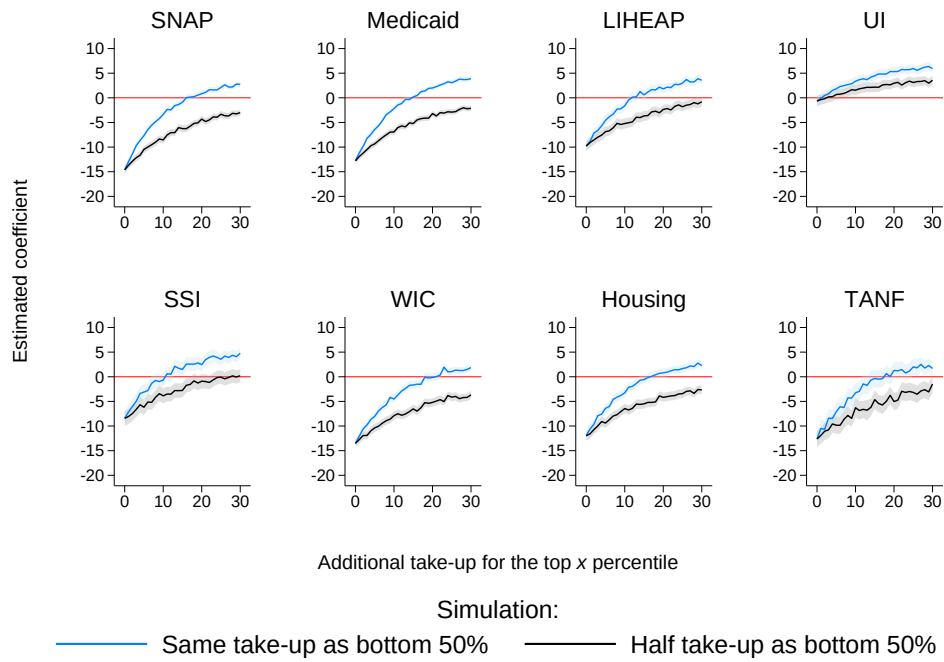


Panel B: Household Rank



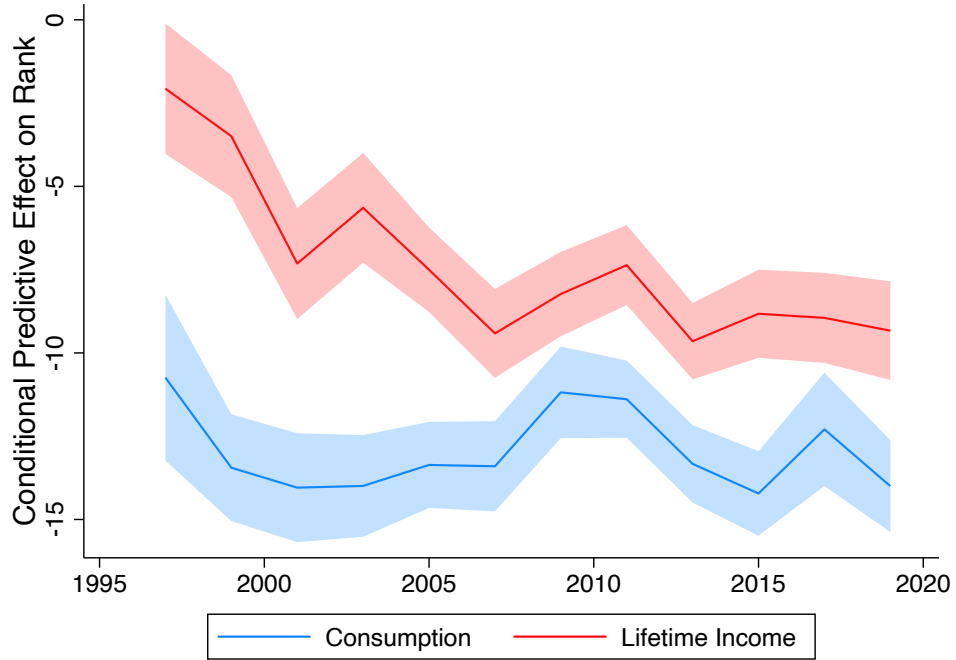
Notes: This figure displays program receipt rates as functions of income percentile ranks, variously for households or individuals, and for current or lifetime income. Section 2 introduces data sources and the measurement of lifetime income. The functions are estimated using a local linear regression with a bandwidth of three percentiles. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands, as in [Chernozhukov et al. \(2013\)](#), with clustering by individual.

Figure A5: Measurement Error Simulations



Notes: This figure displays xxx.

Figure A6: Selection into Transfer Receipt Over Time



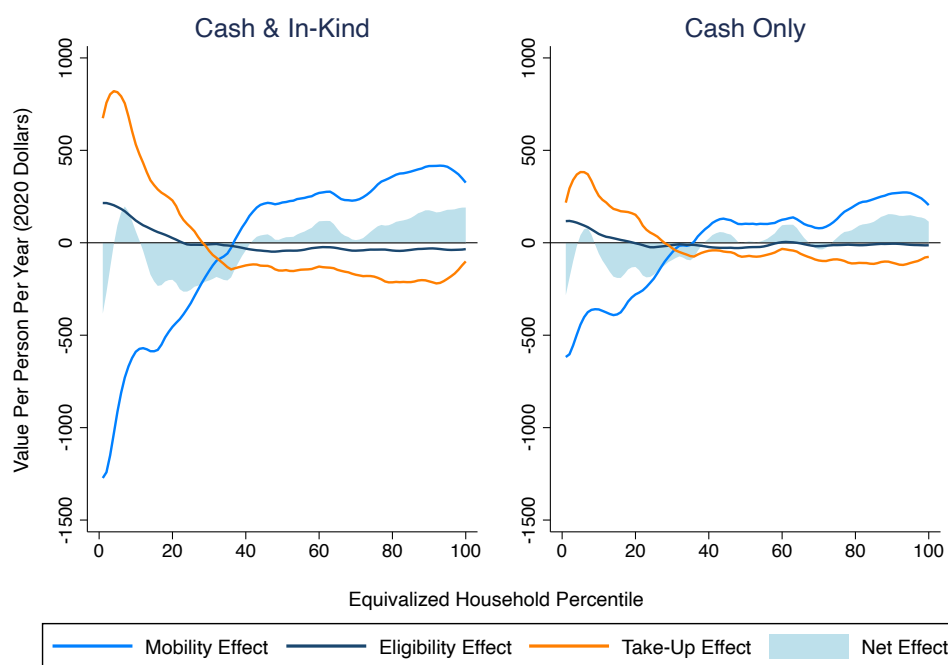
Notes: This figure displays coefficients from the following regression specification:

$$\bar{R}_{its} = \alpha_{cts} + \beta_t D_{its} + f_s(R_{it}) + u_{its},$$

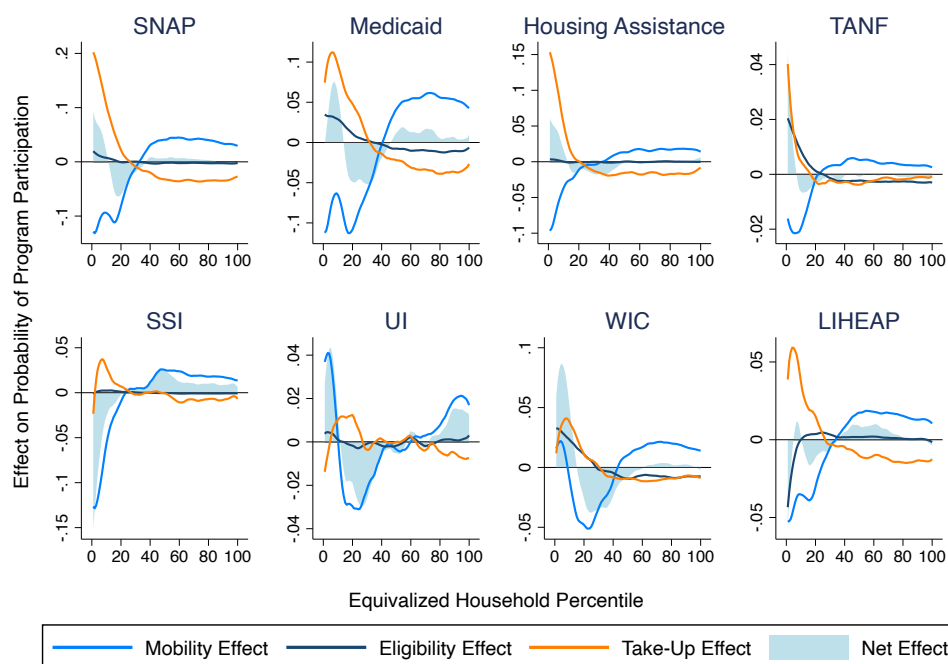
where i denotes households, t denotes years, and s denotes transfer programs. The outcome $\bar{R}_{its}$ is equivalized household consumption rank in the blue line and equivalized household lifetime income rank in red. The data are stacked across programs, so that each individual–year appears eight times, once for each transfer program s . We thus include cohort-year effects α_{cts} specific to each transfer, as well as transfer-specific spline controls $f_s(\cdot)$ for current-income rank R_{it} . The coefficients β_t thus report an average selection effect across transfer programs in a given year t . Shaded regions reflect 95-percent pointwise confidence intervals, with standard errors clustered by individual. All regressions use PSID sample weights but are not otherwise adjust to account for variation in transfer program size.

Figure A7: Decomposition of Difference Between Consumption and Current Incidence

Panel A: Average Total Annual Per-Capita Benefits

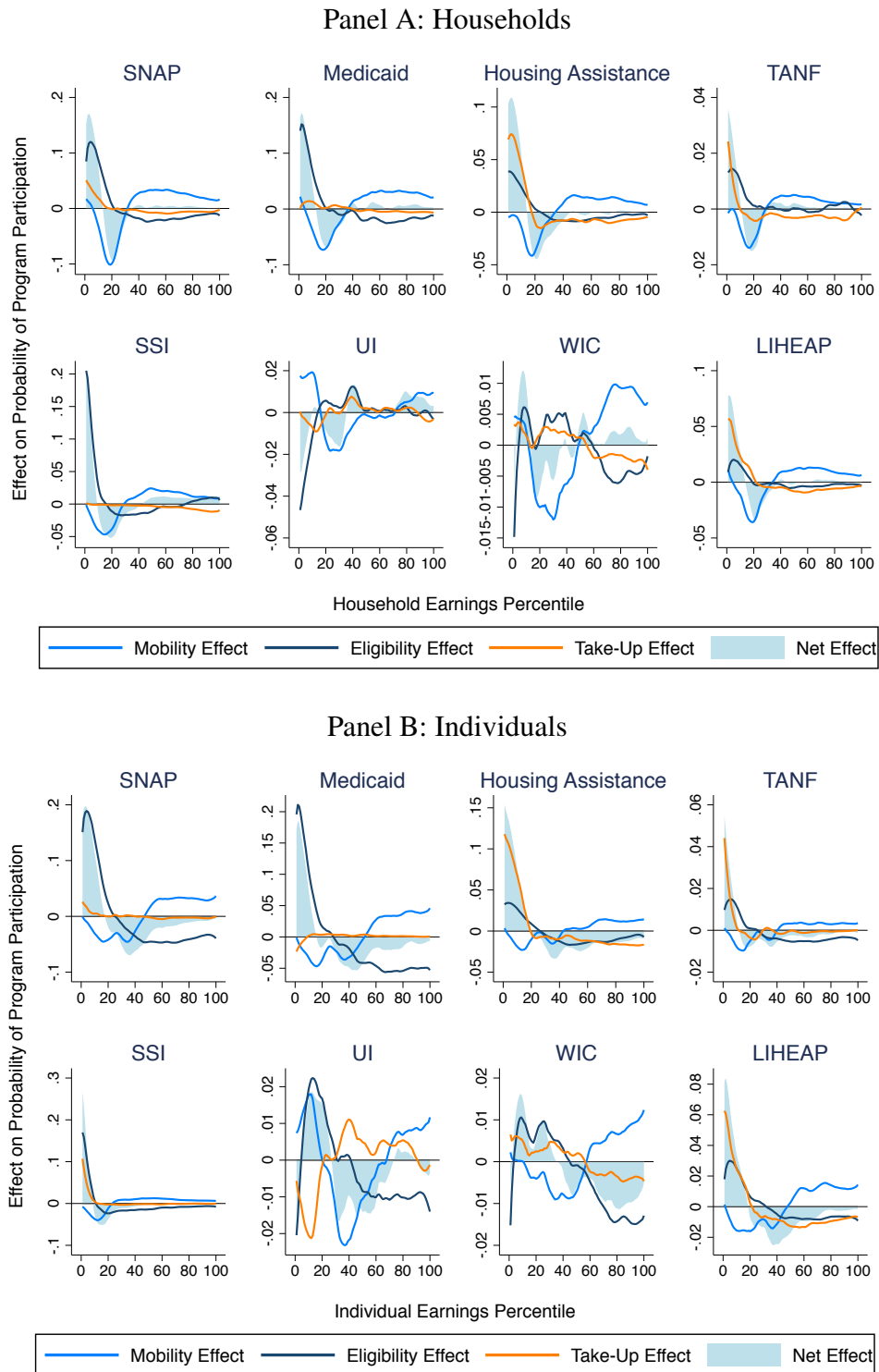


Panel B: Receipt Rate by Program



Notes: This figure displays decompositions of differences between current-income and consumption incidence into mobility, eligibility, and take-up effects. The underlying functions are estimated using local linear regressions with bandwidths of three percentiles.

Figure A8: Decomposition of Difference Between Lifetime and Current Incidence



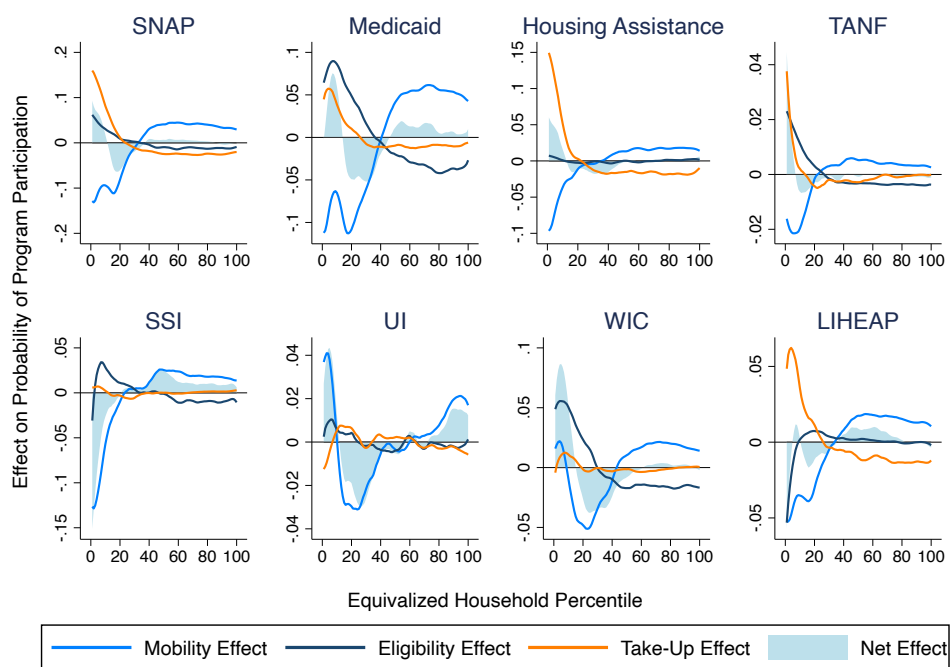
Notes: This figure displays a decomposition of the difference between current and lifetime incidence into mobility and self-selection effects. The original functions are estimated using a local linear regression with a bandwidth of three percentiles.

Figure A9: Decomposition of Difference Between Current-Earnings and Consumption Incidence,
with Adjustment to Simulated Eligibility

Panel A: Average Total Annual Per-Capita Benefits

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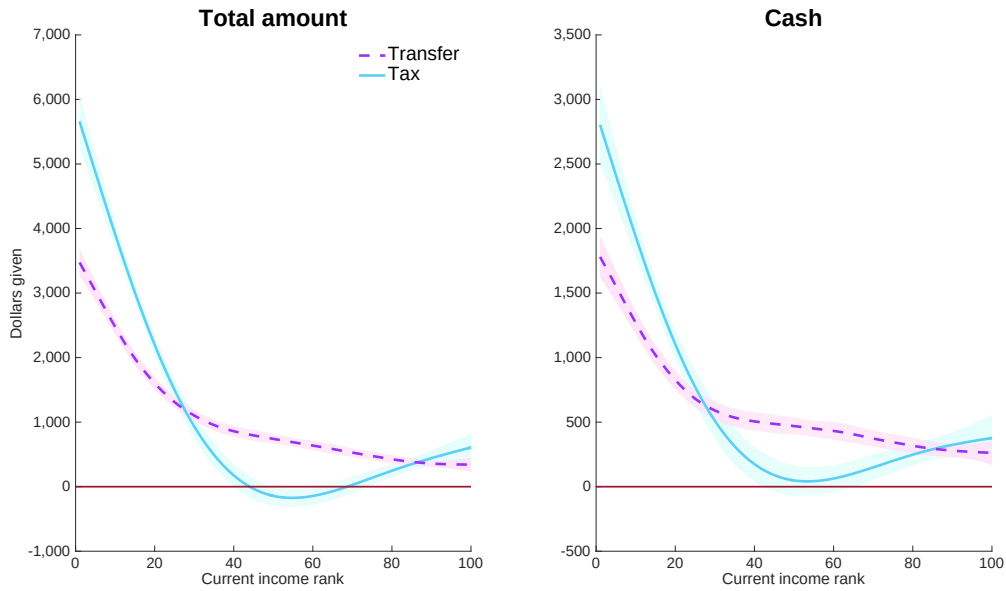
Panel B: Receipt Rate by Program



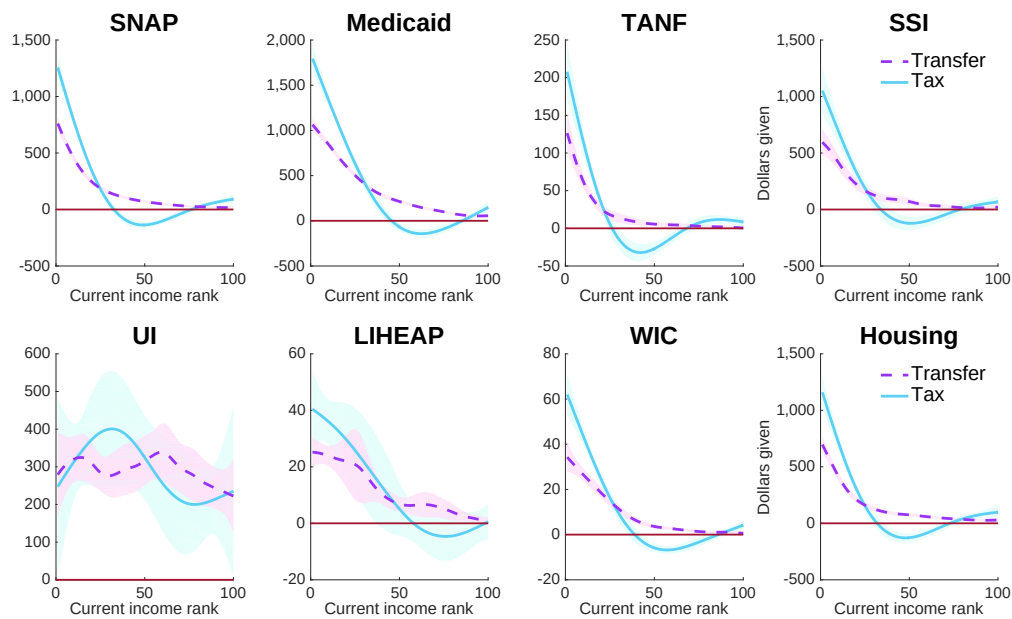
Notes: This figure displays decompositions of differences between current-earnings and consumption incidence into mobility, eligibility, and take-up effects. The underlying functions are estimated using local linear regressions with bandwidths of three percentiles.

Figure A10: Using Income Taxes to Replicate Transfers

Panel A: Average Total Annual Per-Capita Transfer



Panel B: Amount by Program



Notes: This figure displays the average annual transfer payments received by people across the current-income distribution (purple line) and the negative income tax that would replicate the consumption incidence of the transfer program (blue line). Panel A depicts this for all and cash-only transfer programs in total, and Panel B shows the results by program. We explain the construction of replicating tax changes in the text. Shaded regions reflect bootstrapped 95-percent simultaneous confidence bands with clustering by individual.

Table A1: Selection into Transfers, Adjusted for Misreporting of Transfer Receipt

	Baseline		Adjusted for Misreporting	
	Consumption (1)	Lifetime Income (2)	Consumption (3)	Lifetime Income (4)
<i>Panel A: SNAP</i>				
Receives Transfer	-17.6*** (0.6)	-11.1*** (0.6)	-26.4*** (0.8)	-14.3*** (0.9)
<i>Panel B: Medicaid</i>				
Receives Transfer	-14.4*** (0.5)	-7.0*** (0.5)	-23.4*** (0.7)	-12.2*** (0.8)

Notes: This table examines the effect of corrections for misreporting of transfer receipt on estimates of selection into transfers by consumption rank and lifetime-income rank, conditional on current-income rank. The estimating equation is Equation 2. In Columns 3 and 4, we replace reported receipt with the adjusted measures from [Mittag \(2019\)](#) for SNAP and [Davern et al. \(2019\)](#) for Medicaid. All specifications control flexibly for current-income rank using cubic basis splines. Standard errors are clustered by individual. * = $p < 0.10$, ** = $p < 0.05$, *** = $p < 0.01$.

Table A2: Well-Measured Consumption and Transfer Receipt

	SNAP (1)	Medicaid (2)	Housing Assistance (3)	TANF (4)	SSI (5)	UI (6)	WIC (7)	LIHEAP (8)
<i>Panel A: Rent and Owner's Equivalent Rent (1997–2019)</i>								
Receives Transfer	-0.444*** (0.023)	-0.362*** (0.016)	-0.677*** (0.033)	-0.639*** (0.053)	-0.264*** (0.028)	-0.008 (0.018)	-0.389*** (0.021)	-0.359*** (0.034)
<i>Panel B: Vehicle Lease Cost and Equivalent Lease Cost (1999–2019)</i>								
Receives Transfer	-0.234*** (0.011)	-0.253*** (0.008)	-0.109*** (0.017)	-0.361*** (0.025)	-0.048*** (0.012)	0.016 (0.010)	-0.300*** (0.011)	-0.177*** (0.015)
<i>Panel C: Food at Home Expenditure (1999–2019)</i>								
Receives Transfer	-0.530*** (0.015)	-0.244*** (0.011)	-0.257*** (0.020)	-0.508*** (0.035)	-0.174*** (0.020)	-0.008 (0.011)	-0.262*** (0.016)	-0.318*** (0.021)
<i>Panel D: Utility Expenditure (1999–2019)</i>								
Receives Transfer	-0.042** (0.018)	-0.096*** (0.013)	-0.269*** (0.032)	-0.209*** (0.037)	0.012 (0.022)	-0.038*** (0.013)	-0.121*** (0.013)	0.029 (0.022)
<i>Panel E: Gasoline Expenditure (1999–2019)</i>								
Receives Transfer	-0.096*** (0.029)	-0.113*** (0.022)	-0.153*** (0.041)	-0.267*** (0.068)	-0.037 (0.043)	0.094*** (0.021)	-0.107*** (0.029)	-0.215*** (0.038)
Current-Income Spline	X	X	X	X	X	X	X	X

Notes: This table reports estimates of the predictive effect of transfer receipt on (log) levels of reported consumption, conditional on current-income rank. Each panel row is for a different consumption outcome, and each column is for a different transfer. The year ranges in parentheses indicate data coverage for the outcome of interest. All specifications control flexibly for current-income rank using cubic basis splines. Standard errors are clustered by individual. * = $p < 0.10$, ** = $p < 0.05$, *** = $p < 0.01$.

Table A3: Durable-Goods Ownership and Transfer Receipt

	SNAP (1)	Medicaid (2)	Housing Assistance (3)	TANF (4)	SSI (5)	UI (6)	WIC (7)	LIHEAP (8)
<i>Panel A: HH Owns Primary Residence (1997–2019)</i>								
Receives Transfer	-0.164*** (0.010)	-0.068*** (0.009)	-0.453*** (0.008)	-0.177*** (0.015)	-0.067*** (0.013)	0.017 (0.010)	-0.004 (0.010)	-0.058*** (0.017)
<i>Panel B: Number of Rooms in Home (1997–2019)</i>								
Receives Transfer	-0.553*** (0.038)	-0.563*** (0.034)	-0.732*** (0.045)	-0.695*** (0.072)	-0.095* (0.050)	-0.264*** (0.044)	-0.743*** (0.038)	0.006 (0.054)
<i>Panel C: Central Air Conditioning at Home (1997–2009)</i>								
Receives Transfer	-0.037*** (0.012)	-0.028*** (0.010)	0.014 (0.016)	-0.157*** (0.025)	0.012 (0.022)	-0.081*** (0.011)	-0.035*** (0.012)	-0.057*** (0.017)
<i>Panel D: HH Owns a Car (1999–2019)</i>								
Receives Transfer	-0.126*** (0.011)	-0.042*** (0.008)	-0.229*** (0.017)	-0.125*** (0.021)	-0.097*** (0.014)	0.028*** (0.007)	0.029*** (0.009)	-0.039*** (0.014)
<i>Panel D: HH Owns a Computer (2003–2019)</i>								
Receives Transfer	-0.119*** (0.011)	-0.048*** (0.008)	-0.134*** (0.018)	-0.080*** (0.029)	-0.034** (0.014)	-0.007 (0.011)	-0.065*** (0.011)	-0.084*** (0.017)
<i>Panel E: HH Owns a Smartphone (2015–2019)</i>								
Receives Transfer	-0.005 (0.015)	0.005 (0.010)	-0.051** (0.022)	0.064* (0.034)	0.001 (0.012)	0.003 (0.013)	0.056*** (0.011)	-0.054** (0.024)
Current-Income Spline	X	X	X	X	X	X	X	X

Notes: This table reports estimates of the predictive effect of transfer receipt on measures of household durable-goods ownership, conditional on current-income rank. Each panel row is for a different consumption outcome, and each column is for a different transfer. The year ranges in parentheses indicate data coverage for the outcome of interest. All specifications control flexibly for current-income rank using cubic basis splines. Standard errors are clustered by individual. * = $p < 0.10$, ** = $p < 0.05$, *** = $p < 0.01$.

Table A4: Simulated eligibility and receipt rates

Program	(1) P(Receive Simulated Eligible)	(2) P(Receive Not Sim. Elig.)	(3) P(Sim. Elig. Receive)	(4) P(Sim. Elig. Not Receive)
Housing Assistance	0.10	0.02	0.82	0.39
LIHEAP	0.11	0.01	0.97	0.72
Medicaid	0.54	0.09	0.49	0.08
SNAP	0.35	0.04	0.69	0.15
SSI	0.28	0.05	0.12	0.02
TANF	0.07	0.00	0.78	0.11
UI	0.33	0.05	0.25	0.03
WIC	0.44	0.02	0.50	0.03

Notes: xxx.

Table A5: Marginal and average tax rates

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Marginal	Avg. marginal	Marginal	Avg. marginal	Marginal	Avg. marginal	Marginal	Avg. marginal
		10th	0–10th	25th	0–25th	50th	0–50th	99th	0–99th
1. Total	<i>Transfer</i>	0.034	0.139	0.020	0.073	0.004	0.041	0.000	0.022
	<i>Replicating Tax</i>	0.061	0.247	0.051	0.136	0.004	0.081	-0.000	0.040
2. Cash	<i>Transfer</i>	0.018	0.071	0.009	0.037	0.001	0.020	0.000	0.011
	<i>Replicating Tax</i>	0.030	0.122	0.024	0.067	0.001	0.039	-0.000	0.019
3. SNAP	<i>Transfer</i>	0.009	0.045	0.004	0.022	0.001	0.012	-0.000	0.006
	<i>Replicating Tax</i>	0.016	0.067	0.012	0.036	-0.000	0.021	-0.000	0.010
4. Medicaid	<i>Transfer</i>	0.008	0.030	0.008	0.017	0.002	0.011	-0.000	0.006
	<i>Replicating Tax</i>	0.016	0.064	0.017	0.037	0.004	0.024	-0.000	0.012
5. TANF	<i>Transfer</i>	0.002	0.010	0.000	0.004	0.000	0.002	0.000	0.001
	<i>Replicating Tax</i>	0.003	0.013	0.002	0.007	-0.000	0.004	0.000	0.002
6. SSI	<i>Transfer</i>	0.008	0.022	0.004	0.012	0.001	0.007	-0.000	0.004
	<i>Replicating Tax</i>	0.013	0.050	0.011	0.028	0.000	0.016	-0.000	0.008
7. UI	<i>Transfer</i>	-0.001	-0.008	0.002	-0.002	-0.001	-0.001	0.000	-0.001
	<i>Replicating Tax</i>	-0.002	-0.009	-0.001	-0.005	0.003	-0.002	-0.000	-0.001
8. LIHEAP	<i>Transfer</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	<i>Replicating Tax</i>	0.000	0.001	0.000	0.000	0.000	0.000	-0.000	0.000
9. WIC	<i>Transfer</i>	0.000	0.001	0.000	0.001	0.000	0.000	0.000	0.000
	<i>Replicating Tax</i>	0.001	0.003	0.001	0.001	0.000	0.001	-0.000	0.000
10. Housing	<i>Transfer</i>	0.009	0.040	0.003	0.020	0.000	0.011	-0.000	0.005
	<i>Replicating Tax</i>	0.015	0.062	0.011	0.034	-0.001	0.019	-0.000	0.009

B Data Appendix

This appendix first explains measurement details for consumption, lifetime income, and simulated eligibility. Next, it presents the extensions we mention in Section 3. These are the decomposition of differences in incidence into mobility, eligibility, and take-up effects, as well as the inverse approach of finding the income-tax reform that would replicate the consumption or lifetime incidence of transfers.

B.1 Consumption Ranks

We compute households' equivalized consumption ranks using the expenditure data available in the PSID in a given year. Not all consumption categories are available in each year. In particular, we observe expenditures on clothing, furniture, travel, and recreation starting in 2005 and computer expenditures starting in 2017. We observe housing rents (actual and imputed) starting in 1997, and all other expenditures starting in 1999. These expenditure categories are childcare, education, food, health, transportation, and utilities (energy and water starting in 1999, phone/cable/internet starting in 2005).

As noted in Section 2, we follow [Meyer and Sullivan \(2023\)](#) in making two adjustments so that we more closely measure consumption rather than expenditure. Broadly, these adjustments estimate consumption flows from households' two key durable goods, homes and vehicles.

For renters, we take their paid rents as their housing consumption. For owner-occupiers, we obtain imputed rents in several steps. In 2017 and 2019, owner-occupier households were asked "If someone were to rent this (apartment/mobile home/home) today, how much do you think it would rent for per month, unfurnished and without utilities?" We take these values as housing consumption for such households. For all years in our sample period, households who report that their housing is free are asked "How much would it rent for if it were rented?", which we use as their housing consumption. Finally, we construct a mapping from home values to owners' equivalent rents using the cross-sectional relationship in 2017 and 2019 between households' estimates of their home's value and its equivalent rent.²¹

Transportation consumption is constructed as follows. We count any expenditures on gasoline, parking, public transportation, taxis, other transportation towards the household's transportation consumption. Due to PSID data limitations, we also count as consumption any expenditures on vehicles other than the household's three reported primary vehicles. For households that lease any of their three primary vehicles, we count their lease costs toward transportation consumption. For

²¹For households that do not report an exact home value, we use the midpoint of the elicited range. For households who say their homes are worth more than \$400,000, but do not report an exact value, we impute it as the sample mean conditional on exceeding \$400,000 among households who report exact home values.

households that own any of their three primary vehicles, we impute the equivalent lease cost from a hedonic regression.

To estimate this hedonic regression, we restructure our data into a vehicle-level dataset. Households that lease or own a vehicle report the vehicle’s manufacturer (e.g., Toyota), its make (e.g., Lexus), its age at acquisition (year of purchase or lease minus model year), and its “type” (car, pickup/truck, van, utility, or motor home). We estimate Poisson regression models of all two-way interactions of these variables, along with indicator variables for calendar year and the rank (1/2/3) of the vehicle in the household’s list. The outcomes are purchase price or lease cost, winsorizing values at the first and 99th percentiles. We then collapse these predicted values for purchase price and lease cost to the level of manufacturer, make, age, and type. This procedure yields an estimated lease cost equivalent for owned vehicles.²²

B.2 Lifetime Income Ranks

We estimate lifetime income using the PSID’s panel of annual income and demographic characteristics. xxx

Step 1: Estimate lifecycle regression parameters. Letting i index individuals, t index calendar years, and a index age in years, we estimate Poisson regression models of the following form:

$$E[y_{it} | X_{it}] = \exp(\alpha_i \lambda_a + X'_{it} \beta_a),$$

where α_i is an individual fixed effect, λ_a is a calendar-year fixed effect, X_{it} is a matrix of time-varying demographic characteristics, and λ_a and β_a are vectors of age-specific coefficients. The outcome y_{it} is individual income. For individuals with zero income in all observed years, we impute a constant annual income of \$100.²³ The age-specific coefficients are initialized to $\lambda_a = 1$ for all a but will be estimated in an outer loop discussed below.

We make several adjustments before using the regression results to estimate lifetime-income ranks of individuals and households.

Step 2: Shrink fixed effects. First, we apply the empirical Bayes methods in [Morris \(1983\)](#) to shrink the estimated individual fixed effects $\hat{\alpha}_i$ towards a conditional expectation fit from several time-invariant individual characteristics.²⁴ These methods accommodate both unequal individual

²²For missing values, we impute using the cross-sectional relationship between fitted purchase prices and fitted lease values from these Poisson regressions.

²³In an unadjusted Poisson regression, estimates of the individual fixed effects α_i diverge to negative infinity for any individual i who earns $y_{it} = 0$ for all observed periods t . By setting their y_{it} to a very low positive value, we obtain convergent fixed effects and rank these individuals at the bottom of the lifetime-income distribution. Importantly, this procedure does affect our estimates of β , as the fixed effects α_i perfectly explain the income of these individuals.

²⁴Recent applications of these methods in economics include [Chandra et al. \(2016\)](#) and [Sorkin \(2018\)](#). We refer

means and unequal sampling variances in the fixed effects by iteratively re-estimating the extent of true heterogeneity among individuals and the conditional expectation function using weighted least squares. In our baseline specification, we use sex, race, and ethnicity to fit this conditional expectation, but below we consider enriched versions that also use parental education and income. Our results do not change appreciably, which suggests the shrunken individual fixed effects capture reasonably well the permanent component of individual income.

Step 3: Outer loop. Haider and Solon (2006) emphasize that the “error-in-variables” model of lifetime income is misspecified, as the predictive effect of individual fixed effects grows over the lifecycle. To account for this, we estimate the λ_a terms in Equation 1 through the following outer loop. Consider the first loop, in which we have initialized $\lambda_a = 1$ and have shrunken estimates of α_i . We can estimate the following Poisson model:

$$E[y_{it} | X_{it}] = \exp(\hat{\alpha}_i \lambda_a + X'_{it} \beta_a),$$

importantly treating $\{\hat{\alpha}_i\}$ as data rather than as parameters. We then obtain coefficient estimates $\{\hat{\lambda}_a\}$, and with these in hand, we return to step 1 and iterate until convergence of $\{\hat{\alpha}_i, \hat{\lambda}_a\}$. In practice, we find that convergence is fast; three runs of the outer loop are sufficient.

Step 4: Balance the panel. Having estimated the model in Equation 1, we use it to predict income from ages 18 to 65, irrespective of the years in which we observe an individual’s actual income. An individual’s predicted income in year t is $\hat{y}_{it} = \exp(\hat{\alpha}_i^* \hat{\lambda}_a + X'_{it} \hat{\beta}_a)$, where $\hat{\alpha}_i^*$ are the shrunken estimates of the individual fixed effects. Lifetime income are then

$$\bar{y}_i = \sum_t \hat{y}_{it},$$

where the summation over t is for the years $\{\underline{T}_i, \dots, \bar{T}_i\}$ in which individual i is between the ages of 18 and 65. Importantly, however, we do not observe individual characteristics X_{it} in all years and therefore must impute them. In our baseline specification, we assume these characteristics are unchanged from the nearest period of observation, except for age. Below we also test the robustness of our results to imputation strategies which allow these characteristics to evolve.

Step 5: Construct ranks. We define an individual’s lifetime income percentile rank as $\Pr(y \leq \bar{y}_i | c_i = c)$, where $\bar{y}_i$ is their estimated lifetime income and c_i is their birth-year cohort. We define an individual’s current income percentile rank as $\Pr(y \leq y_{it} | c_i = c)$, again ranking individuals each year within their birth cohorts. Appendix A presents figures of our main results when we do

interested readers to their appendices for detailed expositions. One key modification we make to their approach is to use a within-individual Bayesian bootstrap (Rubin, 1981) instead of actual resampling.

not rank current income within cohorts.

We construct current and lifetime household income percentile ranks as follows. Let $j(i, t)$ indicate i 's spouse in year t , and let $h(i, t)$ indicate the household of which i is a part at t . As explained above, current household income is the sum of the head's and spouse's individual current income: $y_{i,t}^h = y_{it} + y_{j(i,t),t}$. Our lifetime concept of household income follows each individual through the sequence of households during their adult life, again using individuals' income fitted from Equation 1 and the subsequent adjustments. That is, the lifetime household income of individual i is

$$\bar{y}_i^h = \sum_t e(\hat{y}_{it}^h) = \sum_t e(\hat{y}_{it} + \hat{y}_{j(i,t),t})$$

where t is again summed over the years in which i is between ages 18 and 65. The function $e(\cdot)$ equalizes household income for differences in household size in each year.²⁵ If we were to restrict our sample to stable households over time (as in, e.g., [Fullerton and Lim Rogers, 1993](#)), our definition of household income would coincide exactly with the natural concept. However, it can also accommodate unstable households, which is obviously important, in a way that is meaningful as a measure of living standards. In particular, important because some individuals are supported by their spouse.

B.3 Simulated Eligibility

Supplemental Nutrition Assistance Program (SNAP).

Medicaid.

Housing Assistance.

Supplemental Security Income (SSI).

Unemployment Insurance (UI).

Women, Infants, and Children (WIC).

Low-Income Heating and Energy Assistance Program (LIHEAP).

²⁵We use a two-parameter scale that distinguishes between the numbers of adults and children in the household: $\lambda_{it} = (N_{it}^{\text{adults}} + \beta \cdot N_{it}^{\text{children}})^\gamma$, setting $\beta = 0.7$ and $\gamma = 0.7$ as parameter values. The parameter $\gamma < 1$ accounts for lesser consumption requirements of children, and the parameter $\gamma < 1$ accounts for household economies of scale in consumption.

B.4 A Decomposition of Distributional Incidence

We have so far argued informally through regressions that take-up, more so than eligibility, explains why transfer receipt identifies those with low consumption given income. We now develop a decomposition, related to [Brewer et al. \(2020\)](#), which more carefully distinguishes between selection via eligibility rules and selective take-up among the eligible.

Mobility Effect. The mobility effect is the component of the difference between incidence with respect to consumption and incidence with respect to income that results from year-to-year income mobility. For instance, because college graduates have significant debt early in their careers but later greatly out-earn non-graduates, the lifetime incidence of student loan forgiveness is less progressive than its incidence with respect to current income (?).

To measure the mobility effect, we estimate the share $q(r)$ of people with consumption rank $\bar{R}_i = r$ who would have received a given transfer in a given year if, counterfactually, transfer receipt were only a function of current-income rank. That is, our mobility-only counterfactual is

$$q(r) = \int \Pr(D_{it} = 1 \mid R_{it}) dF(R_{it} \mid \bar{R}_i = r),$$

where D_{it} indicates transfer receipt and $F(R_{it} \mid \bar{R}_i = r)$ is the conditional distribution of current-income ranks R_{it} at consumption rank r . The mobility effect at rank r is the difference between $q(r)$ and the empirical receipt rate at a current-income rank r , $p(r) = \Pr(D_{it} = 1 \mid R_{it} = 1)$. Intuitively, the mobility effect applies the consumption–income “transition matrix” to the receipt rate by income rank, yielding the counterfactual receipt rates at each consumption rank.

Eligibility Effect. The eligibility effect is the component of the difference in incidence due to eligibility rules that, among low-income households, tag those with low consumption and low lifetime income. For instance, asset tests and categorical eligibility for single parents, people with disabilities, and similar groups tend to target benefits to the persistently poor, not people with merely low current income. By contrast, eligibility rules for contributory social insurance programs such as unemployment insurance operate in the opposite direction, restricting transfers to the persistently poor.

To measure the eligibility effect, we define the receipt rate at consumption rank r under a second counterfactual, $s(r)$. Here the probability of take-up conditional on eligibility is a function of current income alone, whereas we let eligibility depend on both current income and consumption (or current income and lifetime income). This counterfactual receipt rate $s(r)$ thus depends only on eligibility and current-income rank. The difference between these two counterfactuals, $q(r)$ and $s(r)$, is the eligibility effect.

When eligibility can be measured perfectly—that is, $E_i = 0$ implies $D_i = 0$ —our second

counterfactual is defined as

$$s(r) = \int \Pr(D_{it} = 1 | R_{it}, E_{it} = 1) \Pr(E_{it} = 1 | R_{it}, \bar{R}_i = r) dF(R_{it} | \bar{R}_i = r),$$

where E_{it} indicates eligibility. This expression emerges from the following reasoning. The probability $\Pr(D_{it} = 1 | R_{it}, E_{it} = 1)$ is the transfer take-up rate among the eligible at current-income rank R_{it} . The probability $\Pr(E_{it} = 1 | R_{it}, \bar{R}_i = r)$ is the eligibility rate at current-income rank R_{it} and consumption rank r . Multiplying these two probabilities yields a predicted receipt rate for people with current-income rank R_{it} that embeds the desired independence assumption: Given current income, take-up among the eligible is uninformative about consumption or lifetime income.²⁶ Integrating over the conditional distribution of current-income ranks given consumption, $F(R_{it} | \bar{R}_i = r)$, we have a counterfactual receipt rate $s(r)$ at consumption rank r that permits a role for eligibility rules while shutting down selective take-up.

Imperfect measurement of eligibility requires a more complex expression for the same counterfactual concept, which we present in Appendix B.

Take-Up Effect. The take-up effect is the component of the difference in incidence that results from the influence of consumption and lifetime income on take-up rates among the eligible. For instance, procedural complexity (Deshpande and Li, 2019; Gray, 2019), “ordeal” costs (Nichols and Zeckhauser, 1982), and information frictions (Finkelstein and Notowidigdo, 2019) could select for or against low lifetime income and consumption among eligible households with similar current incomes. We define the take-up effect as the residual difference between the counterfactual receipt rate $s(r)$ and the empirical receipt rate at the same consumption rank, $\bar{p}(r) = \Pr(D_{it} = 1 | \bar{R}_i = r)$.

xxx

Eligibility Effect. We need to allow for a probability that people receive the transfer despite being simulated-ineligible, that is, $\Pr(D_{it} = 1 | R_{it}, E_{it} = 0) > 0$. However, the counterfactual answers the same question: What would the distributional impact of a transfer program be if take-up, conditional on eligibility and income, was uninformative about consumption? This counterfactual is

$$s(r) = \int \left[\Pr(D_{it} = 1 | R_{it}, E_{it} = 0) \Pr(E_{it} = 0 | R_{it}, \bar{R}_i = r) + \Pr(D_{it} = 1 | R_{it}, E_{it} = 1) \Pr(E_{it} = 1 | R_{it}, \bar{R}_i = r) \right] dF(R_{it} | \bar{R}_i = r),$$

²⁶To better understand this counterfactual, note that the law of conditional probability implies that: $\Pr(D_{it} = 1 | \bar{R}_i = r) = \int \Pr(D_{it} = 1 | R_{it}, \bar{R}_i = r) dF(R_{it} | \bar{R}_i = r)$. By consequence, the counterfactual receipt rate $s(r)$ equals the empirical receipt rate at a given consumption rank if and only if take-up among eligibles is conditionally independent of consumption given current income at all consumption ranks.

where all terms are as defined above. We use this counterfactual our main analysis. As a robustness check, we also use the earlier counterfactual that requires perfect measurement of eligibility and reclassify simulated-ineligible recipients as eligible.

B.5 Results of Decomposition

This decomposition enables us to interpret differences between the consumption incidence and the income incidence of transfer programs. For instance, why do a greater or smaller share of people at consumption rank r receive a transfer than do people at current-income rank r ? Our decomposition is

$$\bar{p}(r) - p(r) = \underbrace{[\bar{p}(r) - s(r)]}_{\text{take-up effect}} + \underbrace{[s(r) - q(r)]}_{\text{eligibility effect}} + \underbrace{[q(r) - p(r)]}_{\text{mobility effect}}, \quad (6)$$

using the objects defined above. Figure A7 displays decomposition results for the difference between current-income and consumption incidence. Panel A displays results for the average total annual per-capita value of benefits, and Panel B displays results for receipt rates by program.²⁷ The blue, black, and yellow lines plot the mobility, eligibility, and take-up effects, respectively; each represents the difference in consumption incidence with respect to current-income incidence if only this effect were present. The blue shaded region indicates the net effect, equal to the difference between the consumption and current-income incidence at the indicated rank; this region integrates to zero.

Absent eligibility and take-up effects, the incidence of transfer programs would be markedly less progressive at the bottom of the distribution. Consistent with our prior results, we find a central role for selective take-up among eligible households, and a modest role for eligibility rules, in shaping the consumption incidence of transfer programs. UI is the main exception to our results, with little role for either eligibility or take-up effects.

In total over transfer programs, selective take-up increases the average annual value of benefits received by the consumption-poorest people by about \$750 per person relative to a counterfactual in which transfer receipt is a function of current income and eligibility alone. Eligibility rules contribute about \$250 per person at the bottom of the consumption distribution. Together these amounts represent about one quarter of the average transfer per capita per year to consumption-poorest people. Mobility tends to shift the incidence of transfers up the consumption distribution, but bograms, the eligibility and take-up effects largely offset the mobility effect.

The primacy of take-up applies broadly across the eight transfer programs we study. Eligibility

²⁷Appendix A contains supplementary figures that step through the decomposition and probe sensitivity to definitional choices.

rules appear most important in Medicaid, TANF, and WIC, whereas they appear essentially irrelevant or operating in the opposite direction for SNAP, LIHEAP, and housing assistance. While mobility alone would reduce the receipt rate among the consumption poor by about 10 percentage points for SNAP, Medicaid, and housing assistance, selective take-up increases receipt rates in these programs at the bottom of the consumption distribution by 10 to 20 percentage points. These are economically large differences relative to underlying rates of program participation as well as relative to the effects of mobility. In the case of UI, neither eligibility nor take-up is important in shaping incidence.

We test the sensitivity of these results to mismeasurement of eligibility in Appendix Figure A9. Reclassifying simulated-ineligible transfer recipients to be eligible, we find that eligibility rules become somewhat more important for Medicaid and WIC, though take-up remains important for these programs. Our results are, again, most sensitive for SSI. We view our results as indeterminate about the relative importance of eligibility versus take-up in the case of SSI. Even allowing for substantial mismeasurement of eligibility, take-up appears important, especially for SNAP, LIHEAP, and housing assistance.

B.6 Taxes Versus Transfers

This section estimates the change in income taxes that would be necessary to replicate the consumption incidence of various transfer programs.

Approach. Imagine the government were to eliminate a transfer program and replace it with an income-tax reform. We want to find the reform that minimizes the net change in consumption incidence. Due to the eligibility and take-up effects of transfer programs, the “distributionally equivalent” tax reform will differ from the observed current-income incidence of the transfer program.²⁸ If and only if the net effect of eligibility and take-up is zero throughout the consumption distribution—and thus mobility alone shapes the consumption incidence of a transfer—then any taxes and transfers with the same income incidence will also have the same consumption incidence.

Let $S(\tilde{r})$ be the average annual dollar value of transfers at consumption rank $\tilde{r}$, and let $T(r)$ be the average change in income taxes at income rank r . The tax change $T^*(r)$ that would exactly offset the transfer cut throughout the consumption distribution is:

$$T^*(r) = \left\{ T(r) : \int T(r) dF(r|\tilde{r}) = S(\tilde{r}) \text{ for all } \tilde{r} \right\}, \quad (7)$$

where $F(r|\tilde{r})$ is the conditional distribution of consumption ranks $\tilde{r}$ at income rank r . This equation thus implicitly defines the income tax $T^*(r)$ as that which, given the joint distribution of income

²⁸We consider static effects, following the Envelope Theorem (Bradford, 1995).

and consumption, exactly replicates the consumption incidence $S(\tilde{r})$ of a transfer.

Estimation of the replicating tax change $T^*(\tilde{r})$ by Equation 7 is an ill-posed inverse problem (Horowitz, 2014), in the sense that $T^*(\tilde{r})$ is highly sensitive to small changes in $S(r)$, as is common in estimation problems of this form. Following the statistics literature, we use Tikhonov regularization (generalized ridge regression) to reduce this sensitivity. In particular, we penalize the absolute value of the second derivative of $T^*(r)$ according to a parameter λ :

$$T^*(r) = \arg \min_{T(r)} \int \left\| S(\tilde{r}) - \int T(r) dF(r|\tilde{r}) \right\| d\tilde{r} + \lambda \|T''(r)\|, \quad (8)$$

where $\|\cdot\|$ indicates the L^2 norm. This approach smooths fluctuations in the replicating tax change without introducing a bias to its level or slope. In implementation, we discretize the distribution of income and consumption ranks into percentiles, and we select the penalty λ on the numerical second derivative via cross-validation (see Appendix B for further details).

Results. We find that, due to the eligibility and take-up effects, the income taxes needed to replicate the consumption incidence of transfer programs are substantially more progressive in income terms than the transfer programs themselves. Figure A10 depicts, in purple lines, the average annual per-capita transfer payments over the current-income distribution, while the blue lines plot the change in income taxes needed to offset the transfer program in consumption terms. For comparability, we smooth these lines using the same regularization parameter.

To achieve the same consumption incidence as transfers, the income tax system would have to provide \$5,500 per person per year on average to those at the bottom of the current-income distribution, or \$2,000 more than the average transfer payment to this population. We easily reject that the replicating income taxes would be as progressive than the transfer system as a whole. Looking by program, the taxes needed to replicate each transfer program we study would have to be substantially more progressive, with the exceptions of UI and LIHEAP. For instance, the replicating taxes for SNAP would have to be about \$500 larger at the bottom of the income distribution than SNAP itself, or about two-thirds of the average SNAP payment to this population. This result reflects highly selective SNAP take-up among people with low consumption (Figure A7).

Appendix Table A5 presents the implied marginal tax rates at selected percentiles of the current-income distribution and average marginal tax rates up to that percentile. Marginal tax rates for the replicating tax are typically about twice as that of the transfer system. For example, the average marginal tax rate between from the bottom to the tenth percentile of the income distribution is 13.9 percent for all transfers together versus 24.7 percent for the replicating income tax. This pattern applies broadly across programs, with the exception of UI.

Empirical Implementation. We discretize the problem to obtain a numerical solution. We form

100×1 matrices $\mathbf{S}$ which stack the transfers given to each rank. Let $\mathbf{\Delta}$ be the 100×100 transition matrix between current-income rank (columns) and consumption rank (rows). Let $\mathbf{\Gamma}$ be the operator that returns a vector of second differences. Then we estimate:

$$\hat{\mathcal{T}} = \arg \min_{\mathcal{T}} [(\mathbf{S} - \mathbf{\Delta}\mathcal{T})'(\mathbf{S} - \mathbf{\Delta}\mathcal{T}) - \lambda (\mathbf{\Gamma}\mathcal{T})'(\mathbf{\Gamma}\mathcal{T})], \quad (9)$$

where the minimization implicitly employs the L^2 norm.²⁹ Each entry T_j of the 100×1 vector $\mathcal{T}$ represents the tax given to rank $R_i = j$, so $T_j \equiv T(R_i = j)$. Therefore, the vector $\hat{\mathcal{T}}$ represents an estimate of the tax schedule that best implements the redistribution achieved by the transfers.

We also estimate a smoothed transfer schedule:

$$\hat{\mathcal{S}} = \arg \min_{\mathcal{S}} [(\mathbf{S} - \mathcal{S})'(\mathbf{S} - \mathcal{S}) - \lambda (\mathbf{\Gamma}\mathcal{S})'(\mathbf{\Gamma}\mathcal{S})]. \quad (10)$$

We then compare $\hat{\mathcal{S}}$ to $\hat{\mathcal{T}}$: this is the difference in the tax and transfer schedules that achieve the same redistribution in consumption terms.

Cross Validation. For each program, we obtain λ by estimating a k -fold cross-validation procedure on Equation (10), for $k = 10$. We randomly partition the empirical values of $\mathbf{S}$ into 10 groups. We conduct a grid search from $\lambda = 500$ to $\lambda = 2000$. For each λ , we obtain a vector $\hat{\mathcal{S}}$ by estimating Equation 10 on the training sample of $k - 1$ folds. We linearly interpolate for values that are left out. We then estimate the mean-squared error for the leave out sample. We average the mean-squared error across all k folds. For each program, we choose the value of λ that minimizes the mean-squared error. The values of λ are: xxx.

Confidence Intervals. We present 95-percent simultaneous confidence intervals estimated via bootstrap. For each rank, we obtain the standard deviation of bootstrap estimates. We obtain the maximum t statistic within each bootstrap, across ranks. We then use the 95th percentile of the distribution of maximum t statistics, across bootstraps.

B.7 Measurement Error: Simulation

Method. How much measurement error is required to overturn our results? We conduct an adversarial simulation to probe the robustness of Figure 2 to extreme amounts of measurement error. We consider the coefficient γ in Equation 2, which represents the marginal effect of take up of a given transfer program on lifetime rank, controlling for current rank. We simulate measurement error as follows:

²⁹For intuition, observe that with $\mathbf{\Gamma} \equiv \mathbf{I}$, Equation 9 yields a standard ridge regression.

1. Obtain the take-up rate among the bottom 50% of households ranked in consumption terms, $\hat{D} \in [0, 1]$.
2. Assign the top $x\%$ to have some constant $c \in [0, 1]$ times the take-up rate of the bottom 50%: $c\hat{D}$.
3. Estimate Equation 2 using the simulated data.

This exercise generates a large amount of measurement error at the top of the distribution. The take-up rate in the bottom 50% of the current consumption distribution is a natural bound on the take-up rate of the top $x\%$ of the consumption distribution, unless the programs' targeting properties are very perverse. The parameter c governs whether the measurement error is as severe as possible ($c = 1$).

Results. Measurement error would need to be very severe to overturn our results. Figure A5 shows the estimated coefficient $\hat{\gamma}$ as a function of the share of the top of the consumption distribution that has severe measurement error. In black, we present the estimates if the true take-up rate is half the bottom 50%'s take-up rate. The blue lines show the estimates if the true take-up rate is the same as the bottom 50%'s take-up rate. When $x = 0$, the estimates coincide with Figure 2. As long as true take-up at the top is half the poorest's take-up rate, we continue to reject $\gamma = 0$. If true take-up is equal, then we can no longer reject $\gamma = 0$ for $x \geq 15$ or so. These results are logical: if we impute take-up rates that are the same as at the bottom of the distribution for much of the top of the distribution, we no longer find evidence of selection. But as long as measurement error does not exceed half the take-up rate, we decisively reject the null.

B.8 Computation of Inverse-Optimum Weights (Hendren, 2020)

Table 4, Row 5, uses inverse-optimum weights designed to apply a procedure from Hendren (2020). First, we compute the value of the fiscal externality from Equation (4) in Hendren (2020), assuming zero participation elasticity and income effect, the same value for the elasticity of taxable income ($= 0.3$), the implied marginal tax rates from the Heathcote et al. (2017) approximation to the tax schedule, and an estimate of the pareto parameter from the American Communities Survey. These fiscal externalities imply a welfare weight for every individual. We collapse these to percentiles. In the second step, we use the consumption–income transition matrix to solve for welfare weights over consumption that would imply the welfare weights over income. We use Tikhonov regularization with 4-fold cross-validation (Section B.6) to smooth this schedule. We also impose that the consumption welfare weight schedule is nondecreasing.

C Theory Appendix

Throughout recall that since

$$M(z) = P(S(z) \geq \kappa) = \int_0^{S(z)} P(\kappa = k) dk \quad (11)$$

we have

$$M'(z) = S'(z)P(\kappa = S(z)) = S'(z)m(z). \quad (12)$$

C.1 The consumer's problem

To ensure that there is no bunching and that labor supply elasticities are well defined, we assume every consumer of type w chooses their pre-tax income z before their realization of κ is drawn. With the quasi-linearity assumptions, this means that labor supply will be chosen according to expected amount of social program dollars, $M(z)S(z)$ that the consumer expects to accrue.

Each individual maximizes

$$\max_z z - T(z) + \int_0^{S(z)} (S(z) - \kappa) \mu(w, \kappa) d\kappa - v(z/w). \quad (13)$$

Hence the consumer's FOC is

$$0 = 1 - T'(z) + S'(z)M(z) - v'(z/w). \quad (14)$$

Next, we calculate individual elasticities derivatives of labor supply with respect to tax and SNAP changes in terms of primitives. Following [Jacquet and Lehmann \(2014\)](#), we apply perturbations to the tax and snap system about z_0 of the form $\hat{T} = T + \tau(z - z_0) - \nu$ and $\hat{S} = S + \varsigma(z - z_0) - \vartheta$. In each case, the marginal tax (transfer program) rate has increased by τ (ς) but the level has decreased by ν (ϑ).

Now the consumer's problem is

$$\max_z z - T(z) - \tau(z - z_0) - \nu + \int_0^{S(z) + \varsigma(z - z_0) - \vartheta} (S(z) + \varsigma(z - z_0) - \vartheta - \kappa) \mu(w, \kappa) d\kappa - v(z/w). \quad (15)$$

The new FOC is

$$\mathcal{F} = 1 - T'(z) - \tau + (S'(z) + \varsigma) \int_0^{S(z) + \varsigma(z-z_0) - \vartheta} (S'(z) + \varsigma) \mu(w, \kappa) dw - v'(z/w). \quad (16)$$

To use the implicit function theorem, we calculate the derivatives:

$$\mathcal{F}_z |_{\tau=\varsigma=\vartheta=0} = -T''(z) - \frac{v''(z/w_i)}{w_i^2} + 2S''(z)(2S'(z) + \varsigma)M(z) \quad (17)$$

$$\mathcal{F}_\tau |_{\tau=\varsigma=\vartheta=0} = -1 \quad (18)$$

$$\mathcal{F}_\varsigma |_{\tau=\varsigma=\vartheta=0} = 2S'(z)M(z), \quad (19)$$

noting that $z \rightarrow z_0$ as $\tau, \varsigma, \vartheta \rightarrow 0$. Hence by the implicit function theorem we have that

$$\frac{\partial z}{\partial \tau} = \frac{-1}{\mathcal{F}_z} \quad (20)$$

$$\frac{\partial z}{\partial \varsigma} = \frac{2S'(z)M(z)}{\mathcal{F}_z}. \quad (21)$$

And the (compensated) elasticities needed are then defined as normal:

$$\epsilon^z = -\frac{\partial z}{\partial \tau} \Big|_{\tau=0} \frac{1 - T'(z)}{z} \quad (22)$$

$$\epsilon^s = \frac{\partial z}{\partial \varsigma} \Big|_{\varsigma=0} \frac{S'(z)}{z}. \quad (23)$$

C.2 Optimal $T(\cdot)$ and $S(\cdot)$ schedules

Optimal $T(\cdot)$: Suppose the planner increases the tax rate at income z by $d\tau$. Breaking down the effect into fiscal and behavioural responses, we have the following changes to welfare, all written in terms of the marginal value of public funds, λ

1. Direct effect (fiscal and welfare)

$$d\tau \int_{x \geq z} (1 - E_\kappa(g(x, \kappa))) dH(x) = \mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) \quad (24)$$

2. Compensated price effect (effect on tax receipts)

$$d\tau \frac{\partial z}{\partial \tau} \cdot [T'(z)] \cdot h(z) = d\tau \left(-\epsilon^z \cdot z \cdot \frac{T'(z)}{1 - T'(z)} h(z) \right) \quad (25)$$

3. Compensated price effect (effect on social program payments)

$$d\tau \frac{\partial z}{\partial \tau} \cdot [-S'(z)M(z) - S'(z)m(z)S(z)] \cdot h(z) = d\tau \frac{\epsilon^z \cdot z}{1 - T'(z)} [S'(z)M(z) + S'(z)m(z)S(z)] \cdot h(z) \quad (26)$$

So the necessary condition for optimality, that $\Delta W = 0$, is given by:

$$0 = \mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) + \left(-\epsilon^z \cdot z \cdot \frac{T'(z)}{1 - T'(z)} h(z) \right) + \frac{\epsilon^z \cdot z}{1 - T'(z)} [S'(z)M(z) + S'(z)m(z)S(z)] \cdot h(z) \quad (27)$$

$$0 = \mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z))(1 - T'(z)) + T'(z) (-\epsilon^z \cdot z' (z) h(z)) + \epsilon^z \cdot z [S'(z)M(z) + S'(z)m(z)S(z)] \cdot h(z) \quad (28)$$

$$T'(z) [\mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) + \epsilon^z \cdot z' (z) h(z)] = \mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) + \epsilon^z \cdot z [S'(z)M(z) + S'(z)m(z)S(z)] \cdot h(z) \quad (29)$$

$$T'(z) = \frac{\mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z))}{\mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) + \epsilon^z \cdot z h(z)} + \frac{\epsilon^z \cdot z [S'(z)M(z) + S'(z)m(z)S(z)] \cdot h(z)}{\mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) + \epsilon^z \cdot z h(z)} \quad (30)$$

The standard optimal tax rate that would obtain if there were no social program, $S(z) = 0$ can be written as

$$\tau^\dagger = \frac{\mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z))}{\mathbb{E}_{x \geq z} [1 - E_\kappa(g(x, \kappa))] (1 - H(z)) + \epsilon^z \cdot z h(z)}. \quad (31)$$

We can then rewrite the optimal tax rate with a non-zero social program as

$$T'(z) = \tau^\dagger + (1 - \tau^\dagger) S'(z) [M(z) + m(z)S(z)]. \quad (32)$$

To interpret this, consider a negatively sloped social program, $S'(z) < 0$. This affects the optimal marginal tax rate in by weakening incentives to work, since working less delivers more SNAP if marginal or inframarginal. Hence marginal tax rates need to decline somewhat to compensate for this incentive effect.

Optimal $S(\cdot)$:

Proof. Suppose the planner decreases the slope of the social program at income z by $d\varsigma$ so that the schedule is shifted upwards at all incomes to the left of z . Breaking down the effect into fiscal and behavioural responses, we have the following changes to welfare, all written in terms of the marginal value of public funds, λ .

1. Direct effect (fiscal and welfare effects on SNAP inframarginals at each income)

$$d\varsigma E_{x < z} [M(x)(E_{\kappa \leq S(z)}(g(x, \kappa)) - 1)] H(z) \quad (33)$$

2. Direct effect (fiscal effect on SNAP marginals at each income).

$$d\varsigma \mathbb{E}_{x \leq z} [-m(x)S(x)] H(z) \quad (34)$$

where we have assumed that the same welfare weight g_s applies for each of the γ snap-dollars that accrue. This would be exactly correct with quasi-linear utility.

3. Compensated price effect (effect on tax receipts)

$$d\varsigma \frac{\partial z}{\partial \varsigma} \cdot [T'(z)] \cdot h(z) = d\varsigma \left(\xi^s \cdot z \cdot \frac{T'(z)}{S'(z)} h(z) \right) \quad (35)$$

4. Compensated price effect (effect on social program payments)

$$d\varsigma \frac{\partial z}{\partial \varsigma} [-M(z)S'(z) - S'(z)m(z)S(z)] h(z) = d\varsigma \xi^s \frac{z}{S'(z)} \cdot [-M(z)S'(z) - S'(z)m(z)S(z)] h(z) \quad (36)$$

So the necessary condition for optimality, $\Delta W = 0$, becomes :

$$\begin{aligned}
0 &= E_{x < z} [M(x)(E_{\kappa \leq S(z)}(g(x, \kappa)) - 1)]H(z) + \mathbb{E}_{x \leq z} [-m(x)S(x)] H(z) \\
&\quad + \left(\xi^s \cdot z \cdot \frac{T'(z)}{S'(z)} h(z) \right) + \xi^s \frac{z}{S'(z)} \cdot [-M(z)S'(z) - S'(z)m(z)S(z)] h(z)
\end{aligned} \tag{37}$$

$$\begin{aligned}
0 &= E_{x < z} [M(x)(E_{\kappa \leq S(z)}(g(x, \kappa)) - 1)]H(z) + \mathbb{E}_{x \leq z} [-m(x)S(x)] H(z) \\
&\quad + \left(\xi^s \cdot z \cdot \frac{T'(z)}{S'(z)} h(z) \right) + \xi^s z \cdot [-M(z) - m(z)S(z)] h(z)
\end{aligned} \tag{38}$$

$$\begin{aligned}
\xi^s \cdot z \cdot \frac{T'(z)}{S'(z)} h(z) &= E_{x < z} [M(x)(1 - E_{\kappa \leq S(z)}(g(x, \kappa)))]H(z) + \mathbb{E}_{x \leq z} [m(x)S(x)] H(z) \\
&\quad + \xi^s z \cdot [M(z) + m(z)S(z)] h(z)
\end{aligned} \tag{39}$$

$$\begin{aligned}
\frac{1}{S'(z)} &= \frac{1}{\xi^s z T'(z) h(z)} (E_{x < z} [M(x)(1 - E_{\kappa \leq S(z)}(g(x, \kappa)))]H(z) + \mathbb{E}_{x \leq z} [m(x)S(x)] H(z)) \\
&\quad + M(z) + m(z)S(z)
\end{aligned} \tag{40}$$

The first term says that when SNAP is well targeted (i.e. when $E_{\kappa \leq S(z)}g(z, \kappa)$ is large) this pushes in favour of a larger marginal $S'(z)$ rate. The second and third terms say that when the fiscal cost of paying the marginals and inframarginals is high, the marginal $S'(z)$ should be lower. Moreover, when ξ^s is large, (i.e. labor supply expands a lot when $S'(z)$ is increased) this pushes in favour of higher $S'(z)$, as does a high marginal tax rate so that SNAP can compensate for that incentive effect.

C.3 A Nichols and Zeckhauser (1982) efficiency result

Proposition 5.

C.4 Proofs

C.4.1 Proof of Proposition 2

Proof. The positivity of the redistribution within z term follows directly from the definition of within-advantageous selection (definition 1).

It remains to show the positivity of the redistribution between z term under within- and between-advantageous selection. Write

$$f_1(z) = -E_{\kappa}(g(z, \kappa)) \tag{41}$$

and

$$f_2(z) = -(M(z) - E_{z'}(M(z')))) \tag{42}$$

such that the redistribution between z term can be written as

$$\int_z E_\kappa [g(z, \kappa)] \{M(z) - E_{z'} [M(z')]\} h(z) dz = \int_z f_1(z) f_2(z) h(z) dz = E_z [f_2(z) f_2(z)] . \quad (43)$$

By the assumption of between-adverse selection, we have that the distribution of $\kappa \mid z$ increases in z in the FOSD sense. Since $M(z)$ is precisely the CDF of $\kappa \mid z$ evaluated at $S(z)$, immediately we have that $M(z)$ decreases in z and hence $f_1(z)$ increases in z .

By within-adverse selection, $g(z, \kappa)$ decreases in κ , hence $-g(z, \kappa)$ increases in κ . Then again by FOSD, we have that $E_\kappa (-g(z, \kappa)) = f_2(z)$ increases in z .

Since $f_1(z)$ and $f_2(z)$ are both increasing functions of z , it follows from [Schmidt \(2003\)](#) that $Cov(f_1(z), f_2(z)) \geq 0$. Noting that $E_z f_2(z) = 0$ we have

$$E_z [f_1(z) f_2(z)] = E_z [f_1] E_z [f_2(z)] + Cov(f_1, f_2) = Cov(f_1, f_2) \geq 0 \quad (44)$$

as required. $\square$

C.4.2 Proof of Proposition 3

Proof. The redistribution between term is given by $\int_z E_\kappa [g(z, \kappa)] \{M(z) - E_{z'} [M(z')]\} h(z) dz$. For the ‘first dollar’, when initially $S(z) = 0$ for all z , we have $M(z) = 0$ for all z and hence $M(z) - E_{z'} (M(z')) = 0$. Eventually, once $S(z)$ gets large (but still flat) everywhere, we have $M(z) = 1 = E_{z'} (M(z'))$ for all z and hence also $M(z) - E_{z'} (M(z')) = 0$. That the redistribution term is positive elsewhere follows from proposition 2.

The redistribution within term is given by $\int_z M(z) \{E_{\kappa \leq S(z)} [g(z, \kappa)] - E_\kappa [g(z, \kappa)]\} h(z) dz$. At the $S(z) = 0$ for all z we have $M(z) = 0$ for all z and hence the redistribution within term is zero. For large enough flat $S(z)$ we have $E_{\kappa \leq S(z)} [g(z, \kappa)] = E_\kappa [g(z, \kappa)]$ and hence the redistribution within term is again zero. Positivity elsewhere follows from within advantageous selection. $\square$

C.4.3 Proof of Proposition 4

Proof. The labor supply term is given by $\int_z \tau'(z) z \frac{T'(z) - \frac{d}{dz}(S(z)M(z))}{1 - T'(z)} E_\kappa [\epsilon^z] E_{z', \kappa} [g(z', \kappa)] h(z) dz$. By construction, $\tau'(z) = \frac{d}{dz} M(z) < 0$ by the assumption of between advantageous selection. Since $\epsilon^z < 0$ by definition, it remains to prove that $T'(z) - \frac{d}{dz}(S(z)M(z)) > 0$.

The condition of the optimal tax system in section C.2 can be rewritten as

$$\mathbb{E}_{x \geq z} [1 - E_\kappa (g(x, \kappa))] (1 - H(z)) = \left(\epsilon^z \cdot \frac{z}{1 - T'(z)} h(z) \right) \left(T'(z) - \frac{d}{dz} (S(z)M(z)) \right) \quad (45)$$

The LHS is positive and hence so must the RHS be. This requires that $T'(z) - \frac{d}{dz}(S(z)M(z)) > 0$.

Intuitively, this is similar to the 'no negative marginal rates' that obtains in the standard optimal tax model. In that model, negative tax rates mean that you could increase the marginal rate, contracting labor supply and increasing tax revenue collected, clearly contradicting optimality. Here, in order that increasing the tax rate doesn't have the same effect, we need that the change in net revenue collected from a slight decrease in labor supply, $T'(z) - \frac{d}{dz}(S(z)M(z))$ is positive.

□

C.4.4 Proof of Proposition 5

Proof. Goes here

□

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